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1+2+2+3+3+5+9 = 20 m

CONTINUITY AND DIFFERENTIABILITY

$f(x)$	$f'(x)$	$f(x)$	$f'(x)$
x^n	nx^{n-1}	$\sin x$	$\cos x$
k	0	$\cos x$	$-\sin x$
$\frac{1}{x}$	$-\frac{1}{x^2}$	$\tan x$	$\sec^2 x$
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$	$\cot x$	$-\operatorname{cosec}^2 x$
e^x	e^x	$\sec x$	$\sec x \tan x$
a^x	$a^x \log a$	$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$
$\log x$	$\frac{1}{x}$	$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$
x	1	$\cos^{-1} x$	$\frac{-1}{\sqrt{1-x^2}}$
		$\tan^{-1} x$	$\frac{1}{1+x^2}$
		$\cot^{-1} x$	$\frac{-1}{1+x^2}$
		$\sec^{-1} x$	$\frac{1}{x\sqrt{x^2-1}}$
		$\operatorname{cosec}^{-1} x$	$\frac{-1}{x\sqrt{x^2-1}}$

$\frac{d}{dx}(uv) = uv' + vu'$
 $\frac{d}{dx}(uvw) = u'vw + uv'w + uvw'$
 $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$

Composite function is function inside a function -

Eg: $y = \sqrt{\sin \sqrt{\sin x}}$
 $y = e^{\sin^{-1} x}$
 $y = \tan(\sec(\log x))$

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To differentiate composite functions, we follow chain rule, i.e.,

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{du} \cdot \frac{du}{dv} \cdots \frac{dv}{dx}$$

Differentiate the following functions:

1) $y = \sin 2x - 3 \sin x + 4 \cot \sqrt{x}$
 $\frac{dy}{dx} = \cos 2x \cdot 2 - 3 \cos x \cdot 1 - 4 \operatorname{cosec}^2 \sqrt{x} \cdot \frac{1}{2\sqrt{x}}$

2) $y = \log(2x^2 - 3x + 5)$
 $\frac{dy}{dx} = \frac{1}{2x^2 - 3x + 5} \cdot (4x - 3)$

3) $y = \cos(e^{3x})$
 $\frac{dy}{dx} = -\sin(e^{3x}) \cdot e^{3x} \cdot 3$

4) $y = \tan 3x + 4 \operatorname{cosec} x - 5 \sec \sqrt{x}$
 $\frac{dy}{dx} = \sec^2 3x \cdot 3 - 4 \operatorname{cosec} x \cdot \cot x - 5 \frac{\sec \sqrt{x} \tan \sqrt{x}}{2\sqrt{x}}$

5) $y = \log(\sin e^{3x})$
 $\frac{dy}{dx} = \frac{1}{\sin e^{3x}} \cdot \cos e^{3x} \cdot e^{3x} \cdot 3$

6) $y = e^{\log \sin 2x} = \sin 2x$ NOTE: 1) $e^{\log x} = x$
 $\frac{dy}{dx} = 2 \cos 2x$ 2) $a^{\log_a x} = x$

7) $y = e^{\log \cos 2x} = \cos 2x$
 $\frac{dy}{dx} = -2 \sin 2x$

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8] $y = e^{\log \sec 2x} = \sec 2x$ $y' = 2 \sec 2x \tan 2x$	18] $y = e^{\frac{1}{2} \log \tan x}$ $= e^{\log \sqrt{\tan x}} = \sqrt{\tan x}$ $y' = \frac{\sec^2 x}{2 \sqrt{\tan x}}$
9] $y = e^{\log \tan 2x} = \tan 2x$ $y' = 2 \sec^2 2x$	19] $y = e^{\frac{1}{2} \log \sec x} = e^{\log \sqrt{\sec x}}$ $= \sqrt{\sec x}$ $y' = \frac{\sec x \tan x}{2 \sqrt{\sec x}}$
10] $y = e^{\log \cot 2x} = \cot 2x$ $y' = -2 \operatorname{cosec}^2 2x$	20] $y = e^{\frac{1}{2} \log \operatorname{cosec} x}$ $= e^{\log \sqrt{\operatorname{cosec} x}} = \sqrt{\operatorname{cosec} x}$ $y' = \frac{-\operatorname{cosec} x \cot x}{2 \sqrt{\operatorname{cosec} x}}$
11] $y = e^{\log \operatorname{cosec} 2x} = \operatorname{cosec} 2x$ $y' = -2 \operatorname{cosec} 2x \cot 2x$	21] $y = e^{\frac{1}{2} \log \cot x}$ $= e^{\log \sqrt{\cot x}} = \sqrt{\cot x}$ $y' = \frac{-\operatorname{cosec}^2 x}{2 \sqrt{\cot x}}$
12] $y = e^{3 \log x} = e^{\log x^3} = x^3$ $y' = 3x^2$	22] $y = 3^{2 \log_3 x} = 3^{\log_3 x^2} = x^2$ $y' = 2x$
13] $y = e^{4 \log x} = e^{\log x^4} = x^4$ $y' = 4x^3$	23] $y = 4^{3 \log_4 x} = 4^{\log_4 x^3} = x^3$ $y' = 3x^2$
14] $y = e^{5 \log x} = e^{\log x^5} = x^5$ $y' = 5x^4$	24] $y = 5^{4 \log_5 x} = 5^{\log_5 x^4} = x^4$ $y' = 4x^3$
15] $y = e^{7 \log x} = e^{\log x^7} = x^7$ $y' = 7x^6$	25] $y = 6^{7 \log_6 x} = 6^{\log_6 x^7} = x^7$ $y' = 7x^6$
16] $y = e^{3 \log \sin x} = e^{\log \sin^3 x} = \sin^3 x$ $y' = 3 \sin^2 x \cdot \cos x$	
17] $y = e^{4 \log \cos x} = e^{\log \cos^4 x} = \cos^4 x$ $y' = -4 \cos^3 x \cdot \sin x$	

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EXERCISE 5.4

$$1] y = \frac{e^x}{\sin x}$$

$$y' = \frac{\sin x e^x - e^x \cos x}{\sin^2 x} = \frac{e^x (\sin x - \cos x)}{\sin^2 x}$$

$$2] y = e^{\sin^{-1} x}$$

$$y' = \frac{e^{\sin^{-1} x}}{\sqrt{1-x^2}}$$

$$3] y = e^{x^3}$$

$$y' = e^{x^3} \cdot 3x^2$$

$$4] y = \sin(\tan^{-1} e^{-x})$$

$$y' = \frac{\cos(\tan^{-1} e^{-x})}{\sqrt{1+(e^{-x})^2}} \cdot e^{-x} (-1)$$

$$= -\frac{e^{-x} \cos(\tan^{-1} e^{-x})}{\sqrt{1+e^{-2x}}}$$

$$5] y = \log(\cos e^x)$$

$$y' = \frac{-\sin e^x}{\cos e^x} \cdot e^x$$

$$= -e^x (\tan e^x)$$

$$6] y = e^x + e^{x^2} + \dots + e^{x^5}$$

$$y = e^x + e^{x^2} + e^{x^3} + e^{x^4} + e^{x^5}$$

$$y' = e^x + 2xe^{x^2} + 3x^2e^{x^3} + 4x^3e^{x^4} + 5x^4e^{x^5}$$

$$7] y = \sqrt{e^{\sqrt{x}}}$$

$$y' = \frac{e^{\frac{\sqrt{x}}{2}}}{2\sqrt{e^{\sqrt{x}}}} \cdot \frac{1}{2\sqrt{x}} = \frac{1}{2\sqrt{x}}$$

$$8] y = \log(\log x)$$

$$y' = \frac{1}{\log x} \cdot \frac{1}{x} = \frac{1}{x \log x}$$

$$9] y = \frac{\cos x}{\log x}$$

$$y' = \frac{-\log x \sin x - \cos x (\frac{1}{x})}{(\log x)^2}$$

$$10] y = \cos(\log x + e^x)$$

$$y' = -\sin(\log x + e^x) \left(\frac{1}{x} + e^x \right)$$

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EXERCISE 5.2

$$1] \sin y = \sin(x^2 + 5)$$

$$y' = \cos(x^2 + 5) \cdot 2x$$

$$2] y = \cos(\sin x)$$

$$y' = -\sin(\sin x) \cdot \cos x$$

$$3] y = \sin(ax + b)$$

$$y' = \cos(ax + b) \cdot a$$

$$4] a) y = \sec(\tan \sqrt{x})$$

$$y' = \sec(\tan \sqrt{x}) \cdot \tan(\tan \sqrt{x}) \cdot \frac{\sec^2 \sqrt{x}}{2\sqrt{x}}$$

$$8] y = \cos \sqrt{x}$$

$$y' = \frac{-\sin \sqrt{x}}{2\sqrt{x}}$$

$$b) y = \operatorname{cosec}(\cot \sqrt{x})$$

$$y' = -\operatorname{cosec}(\cot \sqrt{x}) \cdot \cot(\cot \sqrt{x}) \cdot \frac{-\operatorname{cosec}^2 \sqrt{x}}{2\sqrt{x}}$$

$$= \operatorname{cosec}(\cot \sqrt{x}) \cdot \cot(\cot \sqrt{x}) \cdot \frac{\operatorname{cosec}^2 \sqrt{x}}{2\sqrt{x}}$$

$$5] y = \frac{\sin(ax+b)}{\cos(cx+d)}$$

$$y' = \frac{\cos(cx+d) \cdot \cos(ax+b) \cdot a + \sin(ax+b) \sin(ax+d) \cdot c}{\cos^2(cx+d)}$$

$$6] y = \cos x^3 \cdot \sin^2(x^5)$$

$$y' = -\sin x^3 \cdot 3x^2 \cdot \sin^2(x^5) + \cos x^3 \cdot 2 \sin(x^5) \cdot \cos(x^5) \cdot 5x^4$$

$$7] y = 2\sqrt{\cot x^2}$$

$$y' = \frac{2(-) \operatorname{cosec}^2(x^2) \cdot 2x}{2\sqrt{\cot x^2}} = \frac{-2x \cdot \operatorname{cosec}^2(x^2)}{\sqrt{\cot x^2}}$$

EXERCISE 5.3

NOTE:

Explicit function: A function of the form $y = f(x)$ is called an explicit function (variables are separated in LHS & RHS)

Implicit function: A function of the form $f(x, y)$ or $f(x, y, z)$ or $f(x, y, z, t)$ are called implicit functions where the variables are inseparable.

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To differentiate implicit function, differentiate the equation as it stands and simplify to get $\frac{dy}{dx}$.

Find $\frac{dy}{dx}$ for the following:

** 1]

$$2x + 3y = \sin x$$

$$2 + 3 \frac{dy}{dx} = \cos x$$

$$\frac{dy}{dx} = \frac{\cos x - 2}{3}$$

$$2) \quad 2x + 3y = \sin y$$

$$2 + 3 \frac{dy}{dx} = \cos y \cdot \frac{dy}{dx}$$

$$\frac{dy}{dx} (3 - \cos y) = -2$$

$$\frac{dy}{dx} = \frac{-2}{3 - \cos y} = \frac{2}{\cos y - 3}$$

3]

$$ax + by^2 = \cos y$$

$$a + b \cdot 2y \cdot \frac{dy}{dx} = -\sin y \cdot \frac{dy}{dx}$$

$$\frac{dy}{dx} (2by + \sin y) = -a$$

$$\frac{dy}{dx} = \frac{-a}{2by + \sin y}$$

$$4) \quad xy + y^2 = \tan x + y$$

$$x \frac{dy}{dx} + y + 2y \cdot \frac{dy}{dx} = \sec^2 x + \frac{dy}{dx}$$

$$= \sec^2 x + \frac{dy}{dx}$$

$$\frac{dy}{dx} (x + 2y - 1) = \sec^2 x - y$$

$$\frac{dy}{dx} = \frac{\sec^2 x - y}{x + 2y - 1}$$

5]

$$x^2 + xy + y^2 = 100$$

$$2x + x \frac{dy}{dx} + y + 2y \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} (x + 2y) = -2x - y$$

$$\frac{dy}{dx} = \frac{-(2x + y)}{x + 2y}$$

$$6) \quad x^3 + x^2y + xy^2 + y^3 = 81$$

$$3x^2 + 2x^2 \frac{dy}{dx} + y^2 + 2xy \cdot \frac{dy}{dx} + y^2 + 3y^2 \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} (x^2 + 2xy + 3y^2) = -3x^2 - 2xy - y^2$$

$$\frac{dy}{dx} = \frac{-(3x^2 + 2xy + y^2)}{x^2 + 2xy + 3y^2}$$

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$$7] \sin^2 y + \cos(xy) = k$$

$$2 \sin y \cdot \cos y \cdot \frac{dy}{dx} - \sin(xy) \left[x \frac{dy}{dx} + y \right] = 0$$

$$\frac{dy}{dx} [2 \sin y \cdot \cos y - x \sin(xy)] = y \sin(xy)$$

$$\frac{dy}{dx} = \frac{y \sin(xy)}{\sin 2y - x \sin(xy)}$$

$$8] \sin^2 x + \cos^2 y = 1$$

$$2 \sin x \cdot \cos x - 2 \cos y \sin y \cdot \frac{dy}{dx} = 0$$

$$\sin 2x - \sin 2y \cdot \frac{dy}{dx} = 0$$

$$\frac{\sin 2x}{\sin 2y} = \frac{dy}{dx}$$

Differentiate the following:

$$*) y = \sin^3 2x$$

$$y' = 3 \sin^2 2x \cdot \cos 2x \cdot 2$$

$$*) y = \sec^2 \left(\frac{a}{x^2} \right)$$

$$y' = 2 \sec \left(\frac{a}{x^2} \right) \cdot \sec \left(\frac{a}{x^2} \right) \cdot \tan \left(\frac{a}{x^2} \right) \cdot \left(-\frac{2a}{x^3} \right)$$

$$*) y = \cos^3 3x$$

$$y' = -2 \cos 3x \cdot \sin 3x \cdot 3$$

$$*) y = \cos^4(\log x)$$

$$y' = 4 \cos^3(\log x) \cdot (-1) \cdot \sin(\log x) \cdot \frac{1}{x}$$

$$*) y = \tan^4 x$$

$$y' = 4 \tan^3 x \cdot \sec^2 x$$

$$*) y = \cot^3 \left(\frac{x}{2} \right)$$

$$y' = 3 \cot^2 \left(\frac{x}{2} \right) \cdot (-1) \cdot \operatorname{cosec}^2 \left(\frac{x}{2} \right) \cdot \left(\frac{1}{2} \right)$$

$$*) y = \cos^2 \sqrt{x}$$

$$y' = 2 \cos \sqrt{x} \cdot (-1) \cdot \sin \sqrt{x} \cdot \frac{1}{2\sqrt{x}}$$

$$*) y = \sec^2 3x$$

$$y' = 2 \sec 3x \cdot \sec 3x \cdot \tan 3x \cdot 3$$

$$= 6 \sec^2 3x \cdot \tan 3x$$

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$$*) y = \operatorname{cosec}^2 6x$$

$$y' = -2 \operatorname{cosec} 6x \cdot \operatorname{cosec} 6x \cdot \cot 6x \cdot 6$$

$$*) y = \tan^3 3x$$

$$y' = 3 \tan^2 3x \cdot \sec^2 3x \cdot 3$$

$$*) y = \log (\sec x + \tan x)$$

$$y' = \frac{\sec x \cdot \tan x + \sec^2 x}{\sec x + \tan x}$$

$$*) y = (1 - x + x^2)^3$$

$$y' = 3(1 - x + x^2)^2 \cdot (-1 + 2x)$$

$$= \frac{\sec x (\tan x + \sec x)}{(\sec x + \tan x)}$$

$$*) y = \frac{1}{3\sqrt{2x+3}} = (2x+3)^{-\frac{1}{3}}$$

$$y' = -\frac{1}{3}(2x+3)^{-\frac{4}{3}} \cdot 2$$

$$*) y = \sqrt{1-x}$$

$$y' = \frac{(-1)}{2\sqrt{1-x}}$$

$$*) y = \log (\sec x)$$

$$y' = \frac{\sec x \cdot \tan x}{\sec x} = \tan x$$

$$*) y = \sqrt{\log x}$$

$$y' = \frac{1}{2\sqrt{\log x} \cdot (x)}$$

$$*) y = \sqrt{\cos x}$$

$$y' = \frac{-\sin x}{2\sqrt{\cos x}}$$

$$*) y = e^{\cos 3x}$$

$$y' = -e^{\cos 3x} \cdot \sin 3x \cdot 3$$

$$*) y = (1 - 3x^4)^{\frac{5}{3}}$$

$$y' = \frac{5}{3} (1 - 3x^4)^{\frac{2}{3}} \cdot (-12x^3)$$

$$*) y = \cos\left(\frac{1}{x^2}\right) = \cos(x^{-2})$$

$$y' = -\sin(x^{-2}) \cdot (-2)x^{-3}$$

$$*) y = \sqrt{a^2 + x^2}$$

$$y' = \frac{2x}{2\sqrt{a^2 + x^2}} = \frac{x}{\sqrt{a^2 + x^2}}$$

$$*) y = \sqrt{1-x^2}$$

$$y' = \frac{-2x}{2\sqrt{1-x^2}} = -\frac{x}{\sqrt{1-x^2}}$$

$$*) y = \operatorname{cosec}^2\left(\frac{1}{x}\right)$$

$$y' = -2 \operatorname{cosec}\left(\frac{1}{x}\right) \left(\operatorname{cosec}\left(\frac{1}{x}\right) \cdot \cot\left(\frac{1}{x}\right) \cdot \left(-\frac{1}{x^2}\right)\right)$$

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*] $y = \tan \sqrt{\cos x}$
 $y' = \frac{\sec^2 \sqrt{\cos x} (-\sin x)}{2\sqrt{\cos x}}$

*] $y = \tan^2(1-e^x)$
 $y' = 2 \tan(1-e^x) \cdot \sec^2(1-e^x) \cdot e^x$

*] $y = \frac{e^{\sin x} \cos x}{e^{\sin x} \sin x + 1}$
 $y' = \frac{e^{\sin x} \cdot \cos x \cdot (-\sin x) + \cos x \cdot e^{\sin x} \cdot \cos x}{e^{2\sin x} + 1}$

*] $y = \log(\sin x) \cdot e^{\sqrt{x}}$
 $y' = \log(\sin x) \cdot \frac{e^{\sqrt{x}}}{2\sqrt{x}} + e^{\sqrt{x}} \cdot \frac{\cos x}{\sin x}$

*] $y = (x^3+1) \sec^2 x$
 $y' = (x^3+1) 2 \sec x \cdot \sec x \cdot \tan x + \sec^2 x \cdot 3x^2$
 $= (x^3+1) 2 \sec^2 x \cdot \tan x + \sec^2 x \cdot 3x^2$

*] $y = e^{ax} \sin bx$
 $y' = e^{ax} b \cos x + \sin bx + a e^{ax}$

*] $y = \sin x^\circ$ (degree can't be differentiated)
 $y = \sin\left(\frac{\pi x}{180}\right)$
 $y' = \cos\left(\frac{\pi x}{180}\right) \cdot \left(\frac{\pi}{180}\right)$

*] $y = \cos x^\circ = \cos\left(\frac{\pi x}{180}\right)$
 $y' = -\sin\left(\frac{\pi x}{180}\right) \cdot \left(\frac{\pi}{180}\right)$

*] $y = \tan x^\circ = \tan\left(\frac{\pi x}{180}\right)$
 $y' = \sec^2\left(\frac{\pi x}{180}\right) \cdot \left(\frac{\pi}{180}\right)$

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*] $y = \log_7(\log x) \quad \left[\log_b a = \frac{\log a}{\log b} \right]$

$$y = \frac{\log(\log x)}{\log 7}$$

$$y' = \frac{1}{\log 7} \cdot \frac{1}{\log x(x)}$$

*] $y = \sin^3 4x \cdot \cos^2 3x$

$$y' = \sin^3 4x \cdot 2 \cos 3x \cdot (-1) \sin 3x \cdot 3 + \cos^2 3x \cdot 3 \sin^2 4x \cdot \cos 4x \cdot 4$$

*] $y = \tan^2 2x \cdot \sec^4 3x$

$$y' = \tan^2 2x \cdot 4 \sec^3 3x \cdot \sec 3x \cdot \tan 3x \cdot 3 + \sec^4 3x \cdot 2 \tan 2x \cdot \sec^2 2x \cdot 2$$

*] $y = \frac{\sqrt{1+x}}{x^2}$

$$y' = \frac{\frac{x^2}{2\sqrt{1+x}} - \sqrt{1+x} \cdot 2x}{x^4}$$

*] $y = \frac{x}{\sqrt{a^2 - x^2}}$

$$y' = \frac{\sqrt{a^2 - x^2} - x(-2x)}{2\sqrt{a^2 - x^2}}$$

$$= \frac{a^2 - x^2 + x^2}{(a^2 - x^2)\sqrt{a^2 - x^2}} = \frac{a^2}{(a^2 - x^2)^{\frac{3}{2}}}$$

*] $y = \sqrt{\frac{1+x}{1-x}}$

$$y' = \frac{\sqrt{1-x}}{2\sqrt{1+x}} \cdot \left[\frac{(1-x) + (1+x)}{(1-x)^2} \right]$$

$$= \frac{\sqrt{1-x}}{2\sqrt{1+x}} \cdot \left[\frac{2}{(1-x)\sqrt{1-x}\sqrt{1+x}} \right] = \frac{1}{(1-x)\sqrt{1-x}}$$

*] $y = \sqrt{\frac{1-\cos x}{1+\cos x}}$

$$y' = \frac{\sqrt{1+\cos x}}{2\sqrt{1-\cos x}} \cdot \left[\frac{(1+\cos x)\sin x - (1-\cos x)\sin x}{(1+\cos x)^2} \right]$$

$$= \frac{\sqrt{1+\cos x}}{2\sqrt{1-\cos x}} \cdot \left[\frac{\sin x (1+\cos x + 1 - \cos x)}{(1+\cos x)\sqrt{1+\cos x}\sqrt{1+\cos x}} \right]$$

$$= \frac{2 \sin x}{2\sqrt{1-\cos^2 x}(1+\cos x)} = \frac{1}{1+\cos x}$$

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$$\begin{aligned}
 *] \quad y &= \sqrt{\frac{1-\sin x}{1+\sin x}} \\
 y' &= \frac{\sqrt{1+\sin x}}{2\sqrt{1-\sin x}} \left[\frac{(1+\sin x)(-\cos x) - (1-\sin x)(\cos x)}{(1+\sin x)^2} \right] \\
 &= \frac{\sqrt{1+\sin x}}{2\sqrt{1-\sin x}} \frac{\cos x(-1-\sin x-1+\sin x)}{(1+\sin x)^2} \\
 &= \frac{\sqrt{1+\sin x}}{2\sqrt{1-\sin x}} \frac{-2\cos x}{(1+\sin x)\sqrt{1+\sin x}} \\
 &= \frac{-\cos x}{\sqrt{1-\sin^2 x}(1+\sin x)} = \frac{-1}{1+\sin x}
 \end{aligned}$$

$$\begin{aligned}
 *] \quad y &= \log_x x = \frac{\log x}{\log x} = 1 \\
 y' &= 0
 \end{aligned}$$

$$\begin{aligned}
 *] \quad y &= \log_x (\log x) = \frac{\log (\log x)}{\log x} \\
 y' &= \frac{\frac{\log x}{x \cdot \log x} - \frac{\log (\log x)}{x}}{(\log x)^2} = \frac{1 - \log (\log x)}{x (\log x)^2}
 \end{aligned}$$

$$\begin{aligned}
 *] \quad y &= \log \frac{a^2-x^2}{a^2+x^2} \quad \text{Show that } y' = \frac{-2a^2x}{a^4-x^4} \\
 y &= \log \left(\frac{a^2-x^2}{a^2+x^2} \right)^{\frac{1}{2}} = \frac{1}{2} \log \left(\frac{a^2-x^2}{a^2+x^2} \right) \\
 &= \frac{1}{2} [\log (a^2-x^2) - \log (a^2+x^2)]
 \end{aligned}$$

$$\begin{aligned}
 y' &= \frac{1}{2} \left[\frac{-2x}{a^2-x^2} - \frac{2x}{a^2+x^2} \right] \\
 &= \frac{-2x}{2} \left[\frac{1}{a^2-x^2} + \frac{1}{a^2+x^2} \right] = \frac{-x(a^2+x^2+a^2-x^2)}{(a^2-x^2)(a^2+x^2)} \\
 &= \frac{-2a^2x}{a^4-x^4}
 \end{aligned}$$

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*] If $y = x + \sqrt{1+x^2}$, prove that $y' = \frac{y}{\sqrt{1+x^2}}$

$$y' = 1 + \frac{2x}{2\sqrt{1+x^2}} = \frac{\sqrt{1+x^2} + x}{\sqrt{1+x^2}} = \frac{y}{\sqrt{1+x^2}}$$

*] $y = \sqrt{x+1} + \sqrt{x-1}$, prove that $4(x^2-1)\left(\frac{dy}{dx}\right)^2 = y^2$

$$y = \sqrt{x+1} + \sqrt{x-1}$$

$$y' = \frac{1}{2\sqrt{x+1}} + \frac{1}{2\sqrt{x-1}}$$

$$= \frac{\sqrt{x-1} + \sqrt{x+1}}{2\sqrt{x^2-1}} = \frac{y}{2\sqrt{x^2-1}}$$

$$\Rightarrow 2\sqrt{x^2-1} \cdot \frac{dy}{dx} = y$$

$$4(x^2-1)\left(\frac{dy}{dx}\right)^2 = y^2 \quad (\text{squaring on both sides})$$

*] If $y = \sqrt{x \log x}$, show that $2y \frac{dy}{dx} = 1 + \log x$

$$y' = \frac{1}{2\sqrt{x \log x}} \cdot \left[x \cdot \frac{1}{x} + \log x \right]$$

$$y' = \frac{1}{2y} [1 + \log x]$$

$$\Rightarrow 2y \frac{dy}{dx} = 1 + \log x$$

*] If $y = (x + \sqrt{a^2+x^2})^n$, show that $(a^2+x^2)y_1' = n^2y^2$

$$y_1' = n(x + \sqrt{a^2+x^2})^{n-1} \cdot \left[1 + \frac{2x}{2\sqrt{a^2+x^2}} \right]$$

$$= n \frac{(x + \sqrt{a^2+x^2})^n}{x + \sqrt{a^2+x^2}} \cdot \left[\frac{x + \sqrt{a^2+x^2}}{\sqrt{a^2+x^2}} \right]$$

$$= \frac{ny}{\sqrt{a^2+x^2}}$$

$$\Rightarrow y_1' \sqrt{a^2+x^2} = ny \quad \Rightarrow (a^2+x^2)y_1' = n^2y^2$$

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*] If $y = \log \left[\frac{a+b \tan x}{a-b \tan x} \right]$. Show that $\frac{dy}{dx} = \frac{2ab \sec^2 x}{a^2 - b^2 \tan^2 x}$

$$\begin{aligned} y' &= \left[\frac{a+b \tan x}{a-b \tan x} \right] \cdot \left[\frac{a-b \tan x (b \sec^2 x) - a+b \tan x (-b \sec^2 x)}{(a-b \tan x)^2} \right] \\ &= \left(\frac{a+b \tan x}{a-b \tan x} \right) \cdot \frac{b \sec^2 x (a-b \tan x + a+b \tan x)}{(a-b \tan x)^2} \\ &= \frac{2ab \sec^2 x}{a^2 - b^2 \tan^2 x} \end{aligned}$$

Types of Differentiation:

Differentiation of inverse trigonometric funⁿ by substitution:EXERCISE 5.3

9] $y = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$ $x = \tan \theta \Rightarrow \theta = \tan^{-1} x$

$$y = \sin^{-1} \left(\frac{2 \tan \theta}{1 + \tan^2 \theta} \right)$$

$$y = \sin^{-1} (\sin 2\theta) = 2\theta = 2 \tan^{-1} x$$

$$y' = \frac{2}{1+x^2}$$

10] $y = \tan^{-1} \left(\frac{3x - x^3}{1 - 3x^2} \right)$, $-\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}$ $x = \tan \theta$
 $\theta = \tan^{-1} x$

$$= \tan^{-1} \left(\frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} \right)$$

$$= \tan^{-1} (\tan 3\theta)$$

$$= 3\theta = 3 \tan^{-1} x$$

$$y' = \frac{3}{1+x^2}$$

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$$\begin{aligned}
 11] \quad y &= \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right), \quad 0 < x < 1 \quad x = \tan \theta \quad \theta = \tan^{-1} x \\
 &= \cos^{-1}\left(\frac{1-\tan^2 \theta}{1+\tan^2 \theta}\right) \\
 &= \cos^{-1}(\cos 2\theta) = 2\theta = 2 \tan^{-1} x \\
 y' &= \frac{2}{1+x^2}
 \end{aligned}$$

$$\begin{aligned}
 12] \quad y &= \sin^{-1}\left(\frac{1-x^2}{1+x^2}\right), \quad 0 < x < 1 \quad x = \tan \theta \quad \theta = \tan^{-1} x \\
 &= \sin^{-1}\left(\frac{1-\tan^2 \theta}{1+\tan^2 \theta}\right) \\
 &= \sin^{-1}(\cos 2\theta) = \sin^{-1}\left[\sin\left(\frac{\pi}{2} - 2\theta\right)\right] \\
 &= \frac{\pi}{2} - 2\theta = \frac{\pi}{2} - 2 \tan^{-1} x \\
 y' &= 0 - \frac{2}{1+x^2} = -\frac{2}{1+x^2}
 \end{aligned}$$

$$\begin{aligned}
 13] \quad y &= \cos^{-1}\left(\frac{2x}{1+x^2}\right), \quad -1 < x < 1 \quad x = \tan \theta \quad \theta = \tan^{-1} x \\
 &= \cos^{-1}\left(\frac{2 \tan \theta}{1+\tan^2 \theta}\right) \\
 &= \cos^{-1}(\sin 2\theta) = \cos^{-1}\left[\cos\left(\frac{\pi}{2} - 2\theta\right)\right] \\
 &= \frac{\pi}{2} - 2\theta = \frac{\pi}{2} - 2 \tan^{-1} x \\
 y' &= 0 - \frac{2}{1+x^2} = -\frac{2}{1+x^2}
 \end{aligned}$$

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$$\begin{aligned}
 14] \quad y &= \sin^{-1}(2x\sqrt{1-x^2}) \quad -\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}} \quad \begin{aligned} x &= \sin \theta \\ \theta &= \sin^{-1} x \end{aligned} \\
 &= \sin^{-1}(2 \sin \theta \sqrt{1-\sin^2 \theta}) \\
 &= \sin^{-1}(2 \sin \theta \cos \theta) \\
 &= \sin^{-1}(\sin 2\theta) = 2\theta = 2 \sin^{-1} x \\
 y' &= \frac{2}{\sqrt{1-x^2}}
 \end{aligned}$$

$$\begin{aligned}
 15] \quad y &= \sec^{-1}\left[\frac{1}{2x^2-1}\right], \quad 0 < x < \frac{1}{\sqrt{2}} \quad \begin{aligned} x &= \cos \theta \\ \theta &= \cos^{-1} x \end{aligned} \\
 &= \sec^{-1}\left[\frac{1}{2\cos^2 \theta - 1}\right] \\
 &= \sec^{-1}\left[\frac{1}{2\cos 2\theta}\right] = \sec^{-1}(\sec 2\theta) = 2\theta = 2 \cos^{-1} x \\
 y' &= \frac{-2}{\sqrt{1-x^2}}
 \end{aligned}$$

Logarithmic differentiation:

- If a function has x or y in power, then logarithm is applied and then differentiated.
- If a function is product of many terms, then application of log simplifies the differentiation.
- If log is not applied for functions with power as x/y , differentiation will be incomplete.

$$\begin{aligned}
 *] \quad y &= x^x \\
 \log y &= x \cdot \log x \\
 \frac{1}{y} \frac{dy}{dx} &= \frac{x}{x} + \log x \\
 \frac{dy}{dx} &= y(1 + \log x) \\
 \frac{dy}{dx} &= x^x(1 + \log x)
 \end{aligned}$$

$$\begin{aligned}
 *] \quad y &= \sin x^x + x^x \\
 y &= u + v \rightarrow (1) \\
 u &= \sin x^x \\
 \log u &= x \log(\sin x) \\
 \frac{1}{u} \cdot \frac{du}{dx} &= \frac{x \cos x}{\sin x} + \log(\sin x) \\
 \frac{du}{dx} &= (\sin x)^x (x \cot x + \log(\sin x))
 \end{aligned}$$

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$$\begin{aligned} \textcircled{1} \Rightarrow \frac{dy}{dx} &= \frac{dv}{dx} + \frac{dv}{dx} \\ &= (\sin x)^x [x \cot x + \log(\sin x)] + x^x (1 + \log x) \end{aligned}$$

EXERCISE 5.5

Eg 30] Differentiate $\left(\frac{(x-3)(x^2+4)}{3x^2+4x+5} \right)$ w.r.t x (Apply log on both side)

$$\log y = \frac{1}{2} [\log(x-3) + \log(x^2+4) - \log(3x^2+4x+5)]$$

$$\frac{dy}{dx} = \frac{y}{2} \left[\frac{1}{x-3} + \frac{2x}{x^2+4} - \frac{6x+4}{3x^2+4x+5} \right]$$

$$= \frac{1}{2} \sqrt{\frac{(x-3)(x^2+4)}{3x^2+4x+5}} \left[\frac{1}{x-3} + \frac{2x}{x^2+4} - \frac{6x+4}{3x^2+4x+5} \right]$$

Eg 31] Differentiate a^x w.r.t x .

$$y = a^x$$

$$\log y = x \log a$$

$$\frac{1}{y} \frac{dy}{dx} = \log a \cdot 1$$

$$\frac{dy}{dx} = y \cdot \log a = a^x \log a$$

Eg 32] $y = x^{\sin x}$, $x > 0$ w.r.t x (Apply log on both sides)

$$\log y = \sin x \cdot \log x$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{\sin x}{x} + \log x \cdot \cos x$$

$$\frac{dy}{dx} = x^{\sin x} \left[\frac{\sin x}{x} + \log x \cdot \cos x \right]$$

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Eg 33]

Find $\frac{dy}{dx}$ if $y^x + x^y + x^x = a^b$ $u + v + w = a^b \rightarrow \text{constant}$

$$\frac{du}{dx} + \frac{dv}{dx} + \frac{dw}{dx} = 0 \rightarrow \text{①}$$

$$w = x^x$$

$$\frac{dw}{dx} = x^x (1 + \log x)$$

$$u = y^x \text{ (Apply log on both sides)}$$

$$\log u = x \log y$$

$$\frac{1}{u} \frac{du}{dx} = \frac{x}{y} \frac{dy}{dx} + \log y$$

$$\frac{du}{dx} = y^x \left(\frac{x}{y} \frac{dy}{dx} + \log y \right)$$

$$\frac{du}{dx} = x y^{x-1} \frac{dy}{dx} + y^x \log y$$

$$v = x^y \text{ (Apply log)}$$

$$\log v = y \log x$$

$$\frac{1}{v} \frac{dv}{dx} = \frac{y}{x} + \log x \frac{dy}{dx}$$

$$\frac{dv}{dx} = x^y \left(\frac{y}{x} + \log x \frac{dy}{dx} \right)$$

$$\frac{dv}{dx} = x^{y-1} y + x^y \log x \frac{dy}{dx}$$

$$\text{①} \Rightarrow x y^{x-1} \frac{dy}{dx} + y^x \log y + x^{y-1} y + x^y \log x \frac{dy}{dx} + x^x (1 + \log x) = 0$$

$$\frac{dy}{dx} [x y^{x-1} + x^y \log x] = -[y^x \log y + x^{y-1} y + x^x (1 + \log x)]$$

$$\therefore \frac{dy}{dx} = - \frac{[y^x \log y + x^{y-1} y + x^x (1 + \log x)]}{[x y^{x-1} + x^y \log x]}$$

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$$1) \quad \cos x \cdot \cos 2x \cdot \cos 3x$$

$$y = \cos x \cdot \cos 2x \cdot \cos 3x \quad (\text{Apply log})$$

$$\log y = \log(\cos x) + \log(\cos 2x) + \log(\cos 3x)$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{-\sin x}{\cos x} + \frac{-\sin 2x(2)}{\cos 2x} + \frac{-\sin 3x(3)}{\cos 3x}$$

$$\frac{dy}{dx} = -\cos x \cdot \cos 2x \cdot \cos 3x (\tan x + 2 \tan 2x + 3 \tan 3x)$$

$$2) \quad y = \sqrt{\frac{(x-1)(x-2)}{(x-3)(x-4)(x-5)}}$$

$$\log y = \frac{1}{2} [\log(x-1) + \log(x-2) - \log(x-3) - \log(x-4) - \log(x-5)]$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{2} \left[\frac{1}{x-1} + \frac{1}{x-2} - \frac{1}{x-3} - \frac{1}{x-4} - \frac{1}{x-5} \right]$$

$$\frac{dy}{dx} = \frac{1}{2} (y) \left[\frac{1}{x-1} + \frac{1}{x-2} - \frac{1}{x-3} - \frac{1}{x-4} - \frac{1}{x-5} \right]$$

$$3) \quad y = (\log x)^{\cos x}$$

$$\log y = \log(\log x)^{\cos x} = \cos x \cdot \log(\log x)$$

$$\frac{1}{y} \frac{dy}{dx} = \left[\frac{\cos x}{x \log x} + \log(\log x) \cdot \sin x \right]$$

$$\frac{dy}{dx} = (\log x)^{\cos x} \left[\frac{\cos x}{(x) \log x} + \log(\log x) \cdot \sin x \right]$$

$$4) \quad y = x^x - 2^{\sin x}$$

$$y = u - v \rightarrow \text{①}$$

$$\frac{dy}{dx} = \frac{du}{dx} - \frac{dv}{dx}$$

$$u = x^x \quad v = 2^{\sin x}$$

$$\frac{du}{dx} = x^x (1 + \log x) \quad \frac{dv}{dx} = 2^{\sin x} \log 2 \cdot \cos x$$

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① $\Rightarrow \frac{dy}{dx} = x^x(1+\log x) - 2^{\sin x} \log 2 \cdot \cos x$

5] $(x+3)^2 \cdot (x+4)^3 \cdot (x+5)^4$
 $y = (x+3)^2 \cdot (x+4)^3 \cdot (x+5)^4$ (Apply log)
 $\log y = 2 \log(x+3) + 3 \log(x+4) + 4 \log(x+5)$
 $\frac{dy}{dx} = y \left[\frac{2}{x+3} + \frac{3}{x+4} + \frac{4}{x+5} \right]$

7] $y = (\log x)^x + x^{\log x}$
 $y = u + v$
 $\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} \rightarrow \text{①}$
 $u = (\log x)^x$
 $\log u = x \log(\log x)$
 $\frac{1}{u} \frac{du}{dx} = \frac{x}{x \cdot \log x} + \log(\log x)$
 $\frac{du}{dx} = u \left[\frac{1}{\log x} + \log(\log x) \right] = (\log x)^x \left[\frac{1}{\log x} + \log(\log x) \right]$
 $v = x^{\log x}$
 $\log v = \log x \cdot \log x = (\log x)^2$
 $\frac{1}{v} \frac{dv}{dx} = \frac{2 \log x}{x}$
 $\frac{dv}{dx} = x^{\log x} \left(\frac{2 \log x}{x} \right)$
 $\text{①} \Rightarrow \frac{dy}{dx} = (\log x)^x \left[\frac{1}{\log x} + \log(\log x) \right] + x^{\log x} \left(\frac{2 \log x}{x} \right)$

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9] $y = x^{\sin x} + \sin x^{\cos x}$

$y = u + v$

$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} \rightarrow \textcircled{1}$

$u = x^{\sin x}$

$\log u = \sin x \cdot \log x$

$\frac{1}{u} \frac{du}{dx} = \frac{\sin x}{x} + \log x \cdot \cos x$

$\frac{du}{dx} = x^{\sin x} \left[\frac{\sin x}{x} + \log x \cdot \cos x \right]$

$v = \sin x^{\cos x}$

$\log v = \cos x \cdot \log(\sin x)$

$\frac{1}{v} \frac{dv}{dx} = \frac{\cos x}{\sin x} \left(\frac{\cos x}{\sin x} \right) + \log(\sin x) (-\sin x)$

$\frac{dv}{dx} = \sin x^{\cos x} \left[\cos x \cdot \cot x - \log(\sin x) \cdot \sin x \right]$

$\textcircled{1} \Rightarrow \frac{dy}{dx} = x^{\sin x} \left[\frac{\sin x}{x} + \log x \cdot \cos x \right] + \sin x^{\cos x} \left[\cos x \cdot \cot x - \log(\sin x) \sin x \right]$

6] $y = \left(x + \frac{1}{x}\right)^x + x^{\left(1 + \frac{1}{x}\right)}$

$y = u + v$

$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} \rightarrow \textcircled{1}$

$u = \left(x + \frac{1}{x}\right)^x$

$\log u = x \cdot \log\left(x + \frac{1}{x}\right)$

$\frac{1}{u} \frac{du}{dx} = \left(\frac{x}{x + \frac{1}{x}}\right) \left(1 - \frac{1}{x^2}\right) + \log\left(x + \frac{1}{x}\right)$

$= \left(\frac{x^2}{x^2 + 1}\right) \left(\frac{x^2 - 1}{x^2}\right) + \log\left(x + \frac{1}{x}\right)$

$\frac{du}{dx} = \left(x + \frac{1}{x}\right)^x \left[\frac{x^2 - 1}{x^2 + 1} + \log\left(x + \frac{1}{x}\right)\right]$

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$$\begin{aligned}
 v &= x^{(1+\frac{1}{x})} \\
 \log v &= (1+\frac{1}{x}) \cdot \log x \\
 \frac{1}{v} \cdot \frac{dv}{dx} &= \frac{(1+\frac{1}{x})}{x} + \log x (0 - \frac{1}{x^2}) \\
 &= \left(\frac{x+1}{x^2}\right) + \left(-\frac{1}{x^2}\right) \log x \\
 &= \frac{x+1-\log x}{x^2} \\
 \frac{dv}{dx} &= x^{(1+\frac{1}{x})} \left[\frac{x+1-\log x}{x^2} \right] \\
 \textcircled{1} \Rightarrow \frac{dy}{dx} &= \left(x+\frac{1}{x}\right)^x \left[\frac{x^2-1}{x^2+1} + \log\left(x+\frac{1}{x}\right) \right] + x^{(1+\frac{1}{x})} \left[\frac{x+1-\log x}{x^2} \right]
 \end{aligned}$$

$$\begin{aligned}
 8] \quad y &= (\sin x)^x + \sin^{-1} \sqrt{x} \\
 y &= u + v \\
 \frac{dy}{dx} &= \frac{du}{dx} + \frac{dv}{dx} \rightarrow \textcircled{1} \\
 u &= (\sin x)^x \\
 \log u &= x \cdot \log(\sin x) \\
 \frac{1}{u} \cdot \frac{du}{dx} &= \frac{x(\cos x)}{\sin x} + \log(\sin x) \cdot 1 \\
 \frac{du}{dx} &= (\sin x)^x [x \cot x + \log(\sin x)] \\
 v &= \sin^{-1} \sqrt{x} \\
 \frac{dv}{dx} &= \frac{1}{\sqrt{1-(\sqrt{x})^2}} \cdot \frac{1}{2\sqrt{x}} = \frac{1}{2\sqrt{x}\sqrt{1-x}} \\
 \frac{dv}{dx} &= \frac{1}{2\sqrt{x-x^2}} \\
 \textcircled{1} \Rightarrow \frac{dy}{dx} &= (\sin x)^x [x \cot x + \log(\sin x)] + \frac{1}{2\sqrt{x-x^2}}
 \end{aligned}$$

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$$10] \quad y = x^{\frac{x \cos x}{x^2-1}} + \frac{x^2+1}{x^2-1}$$

$$y = u + v$$

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} \rightarrow \textcircled{1}$$

$$u = x^{\frac{x \cos x}{x^2-1}}$$

$$\log u = \frac{x \cos x}{x^2-1} \cdot \log x$$

$$\frac{1}{u} \frac{du}{dx} = \frac{x \cos x}{x^2-1} \cdot \frac{1}{x} + \frac{x(-\sin x) \log x}{x^2-1} + \frac{x \cos x}{x^2-1}$$

$$\frac{du}{dx} = x^{\frac{x \cos x}{x^2-1}} \left(\cos x \cdot \log x - x \sin x \cdot \log x + \cos x \right)$$

$$v = \frac{x^2+1}{x^2-1}$$

$$\frac{dv}{dx} = \frac{(x^2-1)(2x) - (x^2+1)(2x)}{(x^2-1)^2} = \frac{2x(x^2-1-x^2-1)}{(x^2-1)^2}$$

$$\frac{dv}{dx} = \frac{2x(-2)}{(x^2-1)^2} = \frac{-4x}{(x^2-1)^2}$$

$$\textcircled{1} \Rightarrow \frac{dy}{dx} = x^{\frac{x \cos x}{x^2-1}} \left(\cos x \cdot \log x - x \sin x \cdot \log x + \cos x \right) - \frac{4x}{(x^2-1)^2}$$

$$11] \quad y = (x \cos x)^x + (x \sin x)^{\frac{1}{x}}$$

$$y = u + v$$

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} \rightarrow \textcircled{1}$$

$$u = (x \cos x)^x$$

$$\log u = x \cdot \log(x \cos x)$$

$$\log u = x \cdot (\log x + \log(\cos x))$$

$$\frac{1}{u} \frac{du}{dx} = x \left(\frac{1}{x} + \frac{-\sin x}{\cos x} \right) + (\log x + \log(\cos x)) \cdot 1$$

$$\frac{du}{dx} = (x \cos x)^x \left(1 - x \tan x + \log(x \cos x) \right)$$

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$$v = (x \sin x)^{\frac{1}{x}}$$

$$\log v = \frac{1}{x} \log(x \sin x)$$

$$\log v = \frac{1}{x} (\log x + \log(\sin x))$$

$$\frac{1}{v} \cdot \frac{dv}{dx} = \frac{1}{x} \left(\frac{1}{x} + \frac{\cos x}{\sin x} \right) + (\log x + \log(\sin x)) \left(-\frac{1}{x^2} \right)$$

$$\frac{1}{v} \cdot \frac{dv}{dx} = \left(\frac{1}{x^2} + \frac{\cot x}{x} - \frac{\log(x \sin x)}{x^2} \right)$$

$$\frac{dv}{dx} = (x \sin x)^{\frac{1}{x}} \left(\frac{1}{x^2} + \frac{\cot x}{x} - \frac{\log(x \sin x)}{x^2} \right)$$

$$\textcircled{1} \Rightarrow \frac{dy}{dx} = (x \cos x)^x [1 - x \tan x + \log(x \cos x)] + (x \sin x)^{\frac{1}{x}} \left(\frac{1}{x^2} + \frac{\cot x}{x} - \frac{\log(x \sin x)}{x^2} \right)$$

Find $\frac{dy}{dx}$ of the functions:

$$12] \quad x^y + y^x = 1$$

$$u + v = 1$$

$$\frac{du}{dx} + \frac{dv}{dx} = 0 \rightarrow \textcircled{1}$$

$$u = x^y$$

$$\log v = y \log x$$

$$\frac{1}{u} \cdot \frac{du}{dx} = \frac{y}{x} + \log x \frac{dy}{dx}$$

$$\frac{du}{dx} = x^y \left[\frac{y}{x} + \log x y' \right]$$

$$u' = y x^{y-1} + x^y \log x y'$$

$$\textcircled{1} \Rightarrow y x^{y-1} + x^y \log x y' + x y^{x-1} y' + y^x \log y = 0$$

$$y' = \frac{-(y x^{y-1} + y^x \log y)}{x^y \log x + x y^{x-1}}$$

$$\log v = x \log y$$

$$\frac{1}{v} \cdot \frac{dv}{dx} = \frac{x}{y} y' + \log y$$

$$\frac{dv}{dx} = y^x \left[\frac{x}{y} y' + \log y \right]$$

$$\frac{dv}{dx} = x y^{x-1} y' + y^x \log y$$

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$$13] \quad x^y = y^x$$

$$y \log x = x \log y$$

$$\frac{y}{x} + \log x \cdot y' = \frac{x}{y} y' + \log y$$

$$y' = \frac{\log y - \frac{y}{x}}{-\frac{x}{y} + \log x} = \frac{(x \log y - y) \cdot y}{(-x + y \log x) \cdot x} = \frac{y(x \log y - y)}{x(y \log x - x)}$$

$$14] \quad (\cos x)^y = (\cos y)^x$$

$$y \log \cos x = x \log \cos y$$

$$\frac{y(-\sin x)}{\cos x} + \log \cos x \cdot y' = \frac{x(-\sin y)}{\cos y} y' + \log(\cos y)$$

$$y' [\log(\cos x) + x \tan x] = \log(\cos y) + y \tan x$$

$$y' = \frac{\log(\cos y) + y \tan x}{\log(\cos x) + x \tan x}$$

$$15] \quad xy = e^{x-y} \text{ Apply log,}$$

$$\log x + \log y = x - y \log e \quad (\because \log e = 1)$$

$$\log x + \log y = x - y \quad (1)$$

$$\frac{1}{x} + \frac{1}{y} \cdot y' = 1 - y'$$

$$y' \left[\frac{1}{y} + 1 \right] = 1 - \frac{1}{x}$$

$$y' \left(\frac{1+y}{y} \right) = \frac{(x-1)}{x}$$

$$y' = \frac{(x-1)y}{(1+y)x}$$

$$16] \text{ Find the derivation of the function given by}$$

$$f(x) = (1+x)(1+x^2)(1+x^4)(1+x^8) \text{ and hence find}$$

$$f'(1)$$

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$$\text{Given: } f(x) = y = (1+x)(1+x^2)(1+x^4)(1+x^8)$$

$$\log y = \log(1+x) + \log(1+x^2) + \log(1+x^4) + \log(1+x^8)$$

$$\frac{1}{y} \cdot y' = \frac{1}{1+x} + \frac{2x}{1+x^2} + \frac{4x^3}{1+x^4} + \frac{8x^7}{1+x^8}$$

$$y' = y \left[\frac{1}{1+x} + \frac{2x}{1+x^2} + \frac{4x^3}{1+x^4} + \frac{8x^7}{1+x^8} \right]$$

$$y' \Big|_{x=1} = 2^4 \left[\frac{1}{2} + \frac{2}{2} + \frac{4}{2} + \frac{8}{2} \right]$$

$$= (16) \left(\frac{15}{2} \right) = 120$$

Parametric Differentiation:

- Generally used variables are x and y . These variables can be intern expressed in other special variables like t, ϕ, ψ etc.
- A pair of equations x and y expressed in special variables form parametric equations. Special variables θ, t, ϕ, ψ are called parameters.
- To differentiate parametric equations, we use

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{dy/dt}{dx/dt} \dots$$

- Differentiation of one function w.r.t another function is also a type of parametric differentiation.

Here, we call the first funⁿ as y and 2nd funⁿ as z and hence required is

$$\frac{dy}{dz} = \frac{dy/d\theta}{dz/d\theta} = \frac{dy/dx}{dz/dx} \dots$$

**

Mis Eg 48]

Differentiate $\sin^2 x$ w.r.t $e^{\cos x}$

$$\begin{array}{l|l} y = \sin^2 x & z = e^{\cos x} \\ \frac{dy}{dx} = 2 \sin x \cdot \cos x & \frac{dz}{dx} = e^{\cos x} \cdot (-\sin x) \end{array}$$

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$$\frac{dy}{dz} = \frac{dy/dx}{dz/dx} = \frac{2 \sin x \cdot \cos x}{-e^{\cos x} \cdot \sin x} = -2 \cos x \cdot e^{-\cos x}$$

Eg 34] Find $\frac{dy}{dx}$ if $x = a \cos \theta$ & $y = a \sin \theta$

$$\frac{dx}{d\theta} = -a \sin \theta$$

$$\frac{dy}{d\theta} = a \cos \theta$$

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{a \cos \theta}{-a \sin \theta} = -\cot \theta$$

Eg 35] Find $\frac{dy}{dx}$ if $x = at^2$, $y = 2at$

$$\frac{dx}{dt} = 2at$$

$$\frac{dy}{dt} = 2a$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2a}{2at} = \frac{1}{t}$$

*** Eg 36] Find $\frac{dy}{dx}$ if $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$

$$\frac{dx}{d\theta} = a[1 + \cos \theta]$$

$$\frac{dy}{d\theta} = a(0 + \sin \theta)$$

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{a \sin \theta}{a(1 + \cos \theta)} = \frac{\sin \theta}{1 + \cos \theta}$$

Eg 37] Find y' if $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ ($\frac{dy}{dx} = -\sqrt[3]{\frac{y}{x}}$)

$$\text{Given: } x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}} \rightarrow (1)$$

$$\left[\begin{array}{l} \text{put } x = a \cos^3 \theta \\ y = a \sin^3 \theta \end{array} \right]$$

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$$\left[\begin{aligned} \text{LHS} &= a^{\frac{2}{3}} \cos^2 \theta + a^{\frac{2}{3}} \sin^2 \theta \\ &= a^{\frac{2}{3}} (\cos^2 \theta + \sin^2 \theta) \\ &= a^{\frac{2}{3}} (1) \\ &= a^{\frac{2}{3}} = \text{RHS} \end{aligned} \right]$$

$$\begin{aligned} x &= a \cos^3 \theta & y &= a \sin^3 \theta \\ \frac{dx}{d\theta} &= a \cdot 3 \cos^2 \theta \cdot (-\sin \theta) & \frac{dy}{d\theta} &= a \cdot 3 \sin^2 \theta \cdot \cos \theta \end{aligned}$$

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{3a \sin^2 \theta \cos \theta}{-3a \cos^2 \theta \sin \theta} = -\frac{\sin \theta}{\cos \theta} \rightarrow (2)$$

$$\begin{aligned} \text{but } x^{\frac{1}{3}} &= a^{\frac{1}{3}} \cos \theta & \text{Ily } y^{\frac{1}{3}} &= a^{\frac{1}{3}} \sin \theta \\ \cos \theta &= \left(\frac{x}{a}\right)^{\frac{1}{3}} & \sin \theta &= \left(\frac{y}{a}\right)^{\frac{1}{3}} \end{aligned}$$

$$(2) \Rightarrow \frac{dy}{dx} = -\frac{\left(\frac{y}{a}\right)^{\frac{1}{3}}}{\left(\frac{x}{a}\right)^{\frac{1}{3}}} = -\left(\frac{y}{x}\right)^{\frac{1}{3}} = -\sqrt[3]{\frac{y}{x}}$$

EXERCISE 5.6

$$\begin{aligned} 1) \quad x &= 2at^2 & y &= at^4 \\ \frac{dx}{dt} &= 4at & \frac{dy}{dt} &= 4at^3 \\ \frac{dy}{dx} &= \frac{dy/dt}{dx/dt} = \frac{4at^3}{4at} = t^2 \end{aligned}$$

$$\begin{aligned} 2) \quad x &= a \cos \theta, \quad y = b \sin \theta \\ \frac{dy}{dx} &= \frac{dy/d\theta}{dx/d\theta} = \frac{-b \sin \theta}{-a \sin \theta} = \frac{b}{a} \end{aligned}$$

$$\begin{aligned} 3) \quad x &= \sin t & y &= \cos 2t \\ \frac{dy}{dx} &= \frac{dy/dt}{dx/dt} = \frac{-2 \sin t}{\cos t} = \frac{-2(2 \sin t \cos t)}{\cos t} = -4 \sin t \end{aligned}$$

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$$4] \quad x = 4t, \quad y = \frac{4}{t}$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = -\frac{4}{t^2} \cdot \frac{1}{4} = -\frac{1}{t^2}$$

$$5] \quad x = \cos \theta - \cos 2\theta, \quad y = \sin \theta - \sin 2\theta$$

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\cos \theta - 2\cos 2\theta}{-\sin \theta + 2\sin 2\theta}$$

$$** 6] \quad x = a(\theta - \sin \theta), \quad y = a(1 + \cos \theta)$$

$$\frac{dx}{d\theta} = a(1 - \cos \theta) \quad \left| \quad \frac{dy}{d\theta} = a(-\sin \theta) \right.$$

$$= a \cdot 2\sin^2 \frac{\theta}{2} \quad \left| \quad = -a \cdot 2\sin \frac{\theta}{2} \cos \frac{\theta}{2} \right.$$

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{-a \cdot 2\sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2a \sin^2 \frac{\theta}{2}} = -\cot \frac{\theta}{2}$$

$$** 8] \quad x = a \left[\cos t + \log \left(\tan \frac{t}{2} \right) \right], \quad y = a \sin t$$

$$\frac{dx}{dt} = a \left\{ -\sin t + \frac{\sec^2(t/2) \cdot \frac{1}{2}}{\tan^2(t/2)} \right\} \quad \frac{dy}{dt} = a \cos t$$

$$= a \left\{ -\sin t + \frac{\cos^2 t/2}{2 \cos^4 t/2 \cdot \sin^2 t/2} \right\}$$

$$= a \left\{ -\sin t + \frac{1}{\sin t} \right\} = a \left\{ \frac{1 - \sin^2 t}{\sin t} \right\} = a \left\{ \frac{\cos^2 t}{\sin t} \right\}$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{a \cos t}{a \frac{\cos^2 t}{\sin t}} \cdot \sin t = \tan t$$

$$** 10] \quad x = a(\cos \theta + \theta \sin \theta), \quad y = a(\sin \theta - \theta \cos \theta)$$

$$\frac{dx}{d\theta} = a \{ -\sin \theta + \theta \cos \theta + \sin \theta \} \quad \left| \quad \frac{dy}{d\theta} = a \{ \cos \theta + \theta \sin \theta - \cos \theta \} \right.$$

$$= a \theta \cos \theta \quad \left| \quad = a \theta \sin \theta \right.$$

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$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{a\theta \sin\theta}{a\theta \cos\theta} = \tan\theta$$

** 11) $x = \sqrt{a^{\sin^{-1}t}}$, $y = \sqrt{a^{\cos^{-1}t}}$ show that $\frac{dy}{dx} = \frac{-y}{x}$

$x = \sqrt{a^{\sin^{-1}t}}$ squaring on both sides,
 $x^2 = a^{\sin^{-1}t}$, $y^2 = a^{\cos^{-1}t} \rightarrow \textcircled{1}$

$$2x \cdot \frac{dx}{dt} = \frac{a^{\sin^{-1}t} \log a}{\sqrt{1-t^2}} \quad \left| \quad 2y \cdot \frac{dy}{dt} = \frac{-a^{\cos^{-1}t} \log a}{\sqrt{1-t^2}} \right.$$

$$\therefore \frac{dx}{dt} = \frac{a^{\sin^{-1}t} \log a}{2x\sqrt{1-t^2}} \quad \left| \quad \frac{dy}{dt} = \frac{-a^{\cos^{-1}t} \log a}{2y\sqrt{1-t^2}} \right.$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-a^{\cos^{-1}t} \log a}{2y\sqrt{1-t^2}} \cdot \frac{2x\sqrt{1-t^2}}{a^{\sin^{-1}t} \log a}$$

using $\textcircled{1}$, $\frac{dy}{dx} = \frac{-y^2 x}{y x^2} = \frac{-y}{x}$

Second order derivative:

If $y = f(x) \rightarrow y$ is a function of 'x'

Differentiating w.r.t 'x'

$$\frac{dy}{dx} = f'(x) = y_1 = y' \rightarrow 1^{\text{st}} \text{ order derivative.}$$

Differentiating again w.r.t 'x'

$$\frac{d^2y}{dx^2} = f''(x) = y_2 = y'' \rightarrow 2^{\text{nd}} \text{ order derivative}$$

Differentiating again w.r.t 'x'

$$\frac{d^3y}{dx^3} = f'''(x) = y_3 = y''' \rightarrow 3^{\text{rd}} \text{ order derivative}$$

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EXERCISE 5.7

1) $y = x^2 + 3x + 2$

$y_1 = 2x + 3$

$y_2 = 2$

2) $y = x^{20}$

$y_1 = 20x^{19}$

$y_2 = (20)(19)x^{18}$

*) $y = x^6$

$y_1 = 6x^5$; $y_2 = 6 \cdot 5 \cdot x^4$; $y_3 = 6 \cdot 5 \cdot 4 \cdot x^3$

$y_4 = 6 \cdot 5 \cdot 4 \cdot 3 \cdot x^2$; $y_5 = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot x$; $y_6 = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 6$

NOTE: $y = x^n \Rightarrow y_n = n!$

3) $y = x \cos x$

$y_1 = -x \sin x + \cos x$

$y_2 = -x \cos x + (-\sin x)1 - \sin x$
 $= -[2 \sin x + x \cos x]$

4) $y = \log x$

$y_1 = \frac{1}{x}$

$y_2 = -\frac{1}{x^2}$

5) $y = x^3 \log x$

$y_1 = \frac{x^3}{x} + \log x \cdot 3x^2 = x^2[1 + 3 \log x]$

$y_2 = \frac{3x^2}{x} + (1 + 3 \log x) \cdot 2x$
 $= x[5 + 6 \log x]$

6) $y = e^x \cdot \sin 5x$

$y_1 = e^x \cdot \cos 5x \cdot 5 + \sin 5x \cdot e^x$

$y_1 = e^x(5 \cos 5x + \sin 5x)$

$y_2 = e^x[-25 \sin 5x + 5 \cos 5x] + (5 \cos 5x + \sin 5x) e^x$
 $= e^x(-24 \sin 5x + 10 \cos 5x)$

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$$\begin{aligned}
 7] \quad y &= e^{6x} \cdot \cos 3x \\
 y_1 &= e^{6x} \cdot (-3 \sin 3x) + (\cos 3x \cdot e^{6x} \cdot 6) \\
 y_1 &= e^{6x} (-3 \sin 3x + 6 \cos 3x) \\
 y_2 &= e^{6x} (-9 \cos 3x - 18 \sin 3x) + (-3 \sin 3x + 6 \cos 3x) \cdot e^{6x} \cdot 6 \\
 y_2 &= e^{6x} (-9 \cos 3x - 18 \sin 3x - 18 \sin 3x + 36 \cos 3x) \\
 y_2 &= e^{6x} (27 \cos 3x - 36 \sin 3x) \\
 y_2 &= 9e^{6x} (3 \cos 3x - 4 \sin 3x)
 \end{aligned}$$

$$\begin{aligned}
 8] \quad y &= \tan^{-1} x \\
 y_1 &= \frac{1}{1+x^2} \\
 y_2 &= \frac{-1}{(1+x^2)^2} \cdot 2x \\
 9] \quad y &= \log(\log x) \\
 y_1 &= \frac{1}{x \cdot \log x} \\
 y_2 &= \frac{-1}{(x \cdot \log x)^2} \cdot \left[\frac{x}{x} + \log x \right] \\
 y_2 &= \frac{-1}{(x \cdot \log x)^2} (1 + \log x)
 \end{aligned}$$

$$\begin{aligned}
 10] \quad y &= \sin(\log x) \\
 y_1 &= \frac{\cos(\log x)}{x} \\
 y_2 &= \frac{x(-\sin(\log x)) - \cos(\log x)}{x^2} \\
 &= \frac{-[\sin(\log x) + \cos(\log x)]}{x^2}
 \end{aligned}$$

$$\begin{aligned}
 ** 11] \quad \text{If } y &= 5 \cos x - 3 \sin x, \text{ Prove that } \frac{d^2 y}{dx^2} + y = 0 \\
 y &= 5 \cos x - 3 \sin x \\
 \frac{dy}{dx} &= -5 \sin x - 3 \cos x \\
 \frac{d^2 y}{dx^2} &= -5 \cos x + 3 \sin x \\
 &= -(5 \cos x - 3 \sin x) \\
 &= -y \\
 \frac{d^2 y}{dx^2} + y &= 0
 \end{aligned}$$

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Eg 39] If $y = A \sin x + B \cos x$. Prove that $\frac{d^2y}{dx^2} + y = 0$

$$y = A \sin x + B \cos x$$

$$\frac{dy}{dx} = A \cos x - B \sin x$$

$$\frac{d^2y}{dx^2} = -A \sin x - B \cos x$$

$$= -(A \sin x + B \cos x) = -y$$

$$\frac{d^2y}{dx^2} + y = 0$$

15] If $y = 500e^{7x} + 600e^{-7x}$. Prove $\frac{d^2y}{dx^2} = 49y$

$$y = 500e^{7x} + 600e^{-7x}$$

$$\frac{dy}{dx} = (500)(7)e^{7x} + (600)(-7)e^{-7x}$$

$$\frac{d^2y}{dx^2} = 500(49)e^{7x} + (600)(49)e^{-7x}$$

$$= 49(500e^{7x} + 600e^{-7x})$$

$$\frac{d^2y}{dx^2} = 49y$$

14] If $y = Ae^{mx} + Be^{nx}$. Show that $\frac{d^2y}{dx^2} - (m+n)\frac{dy}{dx} + mny = 0$

Eg 40] If $y = 3e^{2x} + 2e^{3x}$. Show that $\frac{d^2y}{dx^2} - (2+3)\frac{dy}{dx} + 6y = 0$

14 Ans] $y = Ae^{mx} + Be^{nx}$

$$\frac{dy}{dx} = Ame^{mx} + Bne^{nx}$$

$$\frac{d^2y}{dx^2} = Am^2e^{mx} + Bn^2e^{nx}$$

$$\begin{aligned} \text{LHS} &= \frac{d^2y}{dx^2} - (m+n)\frac{dy}{dx} + mny \\ &= (Am^2e^{mx} + Bn^2e^{nx}) - (m+n)(Ame^{mx} + Bne^{nx}) + mn(Ae^{mx} + Be^{nx}) \end{aligned}$$

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$$\begin{aligned} \text{LHS} &= Am^2e^{mx} + Bn^2e^{nx} - Am^2e^{mx} - Bn^2e^{nx} - Amne^{mx} - Bmne^{nx} \\ &= 0 \\ &= \text{RHS} \end{aligned}$$

Eg 40 Ans] $y = 3e^{2x} + 2e^{3x}$

$$\frac{dy}{dx} = 6e^{2x} + 6e^{3x}$$

$$\frac{d^2y}{dx^2} = 12e^{2x} + 18e^{3x}$$

$$\text{LHS} = \frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y$$

$$= 12e^{2x} + 18e^{3x} - 5[6e^{2x} + 6e^{3x}] + 6[3e^{2x} + 2e^{3x}]$$

$$= 12e^{2x} + 18e^{3x} - 30e^{2x} - 30e^{3x} + 18e^{2x} + 12e^{3x}$$

$$= 12e^{2x} + 18e^{3x} - 12e^{2x} - 18e^{3x}$$

$$= 0$$

$$= \text{RHS}$$

** 13] If $y = 3\cos(\log x) + 4\sin(\log x)$. Show that $x^2y_2 + xy_1 + y = 0$

$$y = 3\cos(\log x) + 4\sin(\log x)$$

$$y_1 = \frac{-3\sin(\log x)}{x} + \frac{4\cos(\log x)}{x}$$

$$xy_1 = -3\sin(\log x) + 4\cos(\log x)$$

$$x^2y_2 + xy_1 = \frac{-3\cos(\log x)}{x} - \frac{4\sin(\log x)}{x}$$

$$x^2y_2 + xy_1 = -[3\cos(\log x) + 4\sin(\log x)]$$

$$x^2y_2 + xy_1 = -y$$

$$x^2y_2 + xy_1 + y = 0$$

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* 17] If $y = (\tan^{-1} x)^2$ show that $(x^2 + 1)^2 y_2 + 2x(x^2 + 1) y_1 = 2$

$$y = (\tan^{-1} x)^2$$

$$y_1 = \frac{2 \tan^{-1} x}{1+x^2}$$

$$(1+x^2) y_1 = 2 \tan^{-1} x$$

$$(1+x^2) y_2 + y_1 (2x) = \frac{2}{1+x^2} = \frac{2}{1+x^2}$$

$$(1+x^2) [(1+x^2) y_2 + y_1 (2x)] = 2$$

$$(1+x^2)^2 y_2 + 2x(x^2+1) y_1 = 2$$

** Eg 4] If $y = \sin^{-1} x$ show that $(1-x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} = 0$

$$y = \sin^{-1} x$$

$$y_1 = \frac{1}{\sqrt{1-x^2}}$$

$$\sqrt{1-x^2} \cdot y_1 = 1 \text{ squaring on both sides,}$$

$$(1-x^2) y_1^2 = 1$$

$$(1-x^2) \cdot 2 y_1 \cdot y_2 + y_1^2 (-2x) = 0$$

Divide by '2y₁'

$$(1-x^2) \cdot y_2 - x y_1 = 0$$

$$(1-x^2) \frac{dy_2}{dx} - x \frac{dy_1}{dx} = 0$$

12] If $y = \cos^{-1} x$ find $\frac{d^2 y}{dx^2}$ in terms of y alone.

$$y = \cos^{-1} x$$

$$\cos y = x$$

$$\cos y = x$$

$$-\sin y \cdot \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = -\frac{1}{\sin y} = -\operatorname{cosec} y$$

$$\frac{d^2 y}{dx^2} = (-)(-) \operatorname{cosec} y \cdot \cot y \cdot \frac{dy}{dx}$$

$$= \operatorname{cosec} y \cdot \cot y (-\operatorname{cosec} y)$$

$$= -\operatorname{cosec}^2 y \cdot \cot y$$

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16] If $e^y(x+1) = 1$. Show that (a) $\frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2$ (b) $\frac{dy}{dx} = -e^y$

(a) $e^y(x+1) = 1 \rightarrow \text{①}$

$$e^y + (x+1) \cdot e^y \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-e^y}{e^y(x+1)} = \frac{-1}{x+1}$$

$$\frac{d^2y}{dx^2} = (-1) \cdot \frac{1}{(x+1)^2} = \left(\frac{-1}{x+1}\right)^2$$

$$\Rightarrow \frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2$$

(b) $\frac{dy}{dx} = \frac{-e^y}{e^y(x+1)} = \frac{-e^y}{(1)}$ [using ①]

MisEx 16] If $\cos y = x \cos(a+y)$, $\cos a \neq \pm 1$, Prove $\frac{dy}{dx} = \frac{\cos^2(a+y)}{\sin a}$

Given: $\cos y = x \cos(a+y)$

$$\Rightarrow x = \frac{\cos y}{\cos(a+y)}, \text{ differentiating w.r.t } x$$

$$\frac{dx}{dy} = \frac{\cos(a+y)(-\sin y) + \cos y \sin(a+y)}{\cos^2(a+y)}$$

$$\frac{dx}{dy} = \frac{\sin(a+y-y)}{\cos^2(a+y)} = \frac{\sin a}{\cos^2(a+y)}$$

$$\therefore \frac{dy}{dx} = \frac{\cos^2(a+y)}{\sin a}$$

MisEx 17] If $x = a(\cos t + t \sin t)$, $y = a(\sin t - t \cos t)$ find $\frac{d^2y}{dx^2}$

$$\frac{dx}{dt} = a[-\sin t + t \cos t + \sin t] \quad \left| \quad \frac{dy}{dt} = a[\cos t + t \sin t - \cos t] \right.$$

$$= at \cos t \rightarrow \text{①} \quad \left| \quad = at \sin t \right.$$

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$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{dt \sin t}{dt \cos t} = \tan t$$

$$\begin{aligned} \frac{d^2y}{dx^2} &= \sec^2 t \cdot \frac{dt}{dx} \\ &= \frac{\sec^2 t}{dt \cos t} \quad [\text{using } \textcircled{1}] \end{aligned}$$

$$\frac{d^2y}{dx^2} = \frac{\sec^3 t}{dt}$$

Mis Ex 23] If $y = e^{\cos^{-1}x}$; $-1 \leq x \leq 1$. Show that $(1-x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} - a^2 = 0$

$$\begin{aligned} y &= e^{\cos^{-1}x} \\ \frac{dy}{dx} &= e^{\cos^{-1}x} \cdot \frac{(-a)}{\sqrt{1-x^2}} = -\frac{ay}{\sqrt{1-x^2}} \end{aligned}$$

$$\sqrt{1-x^2} \frac{dy}{dx} = -ay \quad (\text{squaring on both sides})$$

$$(1-x^2) \left(\frac{dy}{dx} \right)^2 = a^2 y^2$$

$$(1-x^2) \cdot 2y_1 \cdot y_2 + y_1^2 (-2x) = a^2 \cdot 2y \cdot y_1$$

Divide by $2y_1$

$$(1-x^2) \cdot y_2 + y_1(-x) = a^2 y$$

$$(1-x^2) y_2 - y_1 x = a^2 y$$

$$(1-x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} - a^2 y = 0$$

Mis Ex 1] $y = (3x^2 - 9x + 5)^9$
 $y_1 = 9(3x^2 - 9x + 5)^8 \cdot (6x - 9)$

Mis Ex 2] $y = \sin^3 x + \cos^6 x$
 $y_1 = 3\sin^2 x \cdot \cos x - 6\cos^5 x \cdot \sin x$
 $= 3\sin x \cos x (\sin x - 2\cos^4 x)$

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Mis Ex 3] $y = (5x)^{3\cos 2x}$
 power has 'x', hence apply log.
 $\log y = 3\cos 2x \cdot \log 5x$
 $\frac{1}{y} \frac{dy}{dx} = \frac{3\cos 2x}{5x} \cdot 5 + \log 5x \cdot 3(-2\sin 2x)$
 $\frac{dy}{dx} = y \left[\frac{3\cos 2x}{x} - 6\sin 2x \cdot \log 5x \right]$

Mis Ex 4] $y = \sin^{-1}(x\sqrt{x}) = \sin^{-1}(x^{3/2})$
 $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^3}} \cdot \frac{3}{2} \cdot x^{1/2}$

Mis Ex 5] $y = \cos^{-1}\left(\frac{x}{\sqrt{2x+1}}\right), -2 < x < 2$
 $\frac{dy}{dx} = \frac{1}{(\sqrt{2x+1})^2} \cdot \frac{-1}{\sqrt{1-(\frac{x}{\sqrt{2x+1}})^2}} \cdot \frac{1}{2} \cdot \cos^{-1}\left(\frac{x}{\sqrt{2x+1}}\right) \cdot \frac{x}{\sqrt{2x+1}}$

Mis Ex 6] $y = \cot^{-1}\left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}}\right), 0 < x < \pi$
 $y = \cot^{-1}\left(\frac{\cos \frac{x}{2} + \sin \frac{x}{2} + \cos \frac{x}{2} - \sin \frac{x}{2}}{\cos \frac{x}{2} + \sin \frac{x}{2} - \cos \frac{x}{2} + \sin \frac{x}{2}}\right)$
 $= \cot^{-1}\left(\frac{2\cos \frac{x}{2}}{2\sin \frac{x}{2}}\right) = \cot^{-1}\left(\cot \frac{x}{2}\right) = \frac{x}{2}$
 $\frac{dy}{dx} = \frac{1}{2}$

Mis Ex 7] $y = \log x^{\log x}$ power has x, apply log on both sides,
 $\log y = \log x \cdot \log(\log x)$

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$$\frac{1}{y} \frac{dy}{dx} = \log x \cdot \frac{1}{x \log x} + \frac{\log(\log x)}{x}$$

$$\frac{dy}{dx} = \frac{\log x}{x} \left[1 + \log(\log x) \right]$$

Mis Ex 8) $y = \cos(a \cos x + b \sin x)$ for some constant a, b .

$$\frac{dy}{dx} = -\sin(a \cos x + b \sin x) \cdot (-a \sin x + b \cos x)$$

Mis Ex 9) $y = (\sin x - \cos x)^{\sin x - \cos x}$, $\frac{\pi}{4} < x < \frac{3\pi}{4}$

$$\log y = \sin x - \cos x \cdot \log(\sin x - \cos x)$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{\sin x - \cos x}{\sin x - \cos x} (\cos x + \sin x) + \log(\sin x - \cos x) (\cos x + \sin x)$$

$$\frac{dy}{dx} = y (\cos x + \sin x) [1 + \log(\sin x - \cos x)]$$

Mis Ex 10) $y = x^x + x^a + a^x + a^a$ for some fixed $a > 0, x > 0$

$$\frac{dy}{dx} = x^x(1 + \log x) + ax^{a-1} + a^x \log a + 0$$

$$u = x^x \text{ (Apply log)}$$

$$\log v = x \log x$$

$$\frac{1}{u} \frac{du}{dx} = \frac{x}{x} + \log x$$

$$\frac{du}{dx} = x^x(1 + \log x)$$

Mis Ex 11) $y = x^{x^2-3} + (x-3)^{x^2}$

$$y = u + v$$

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} \rightarrow \textcircled{1}$$

$$u = x^{x^2-3}$$

$$\log u = x^2 - 3 \cdot \log x$$

$$\frac{1}{u} \frac{du}{dx} = \frac{x^2-3}{x} + \log x (2x)$$

$$v = (x-3)^{x^2}$$

$$\log v = x^2 \log(x-3)$$

$$\frac{1}{v} \frac{dv}{dx} = \frac{x^2}{x-3} + \log(x-3) (2x)$$

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$$\textcircled{1} \Rightarrow \frac{dy}{dx} = \frac{x^{x-3}}{x} \left[\frac{x^2-3}{x} + \log x (2x) \right] + (x-3)^{x-1} \left[\frac{x^2}{x-3} + \log(x-3)(2x) \right]$$

MisEx12] Find $\frac{dy}{dx}$ if $y = 12(1 - \cos t)$, $x = 10(t - \sin t)$, $-\frac{\pi}{2} < t < \frac{\pi}{2}$

$$\frac{dy}{dt} = 12(0 + \sin t) + (1 - \cos t) \cdot 0 \quad \left| \quad \frac{dx}{dt} = 10(1 - \cos t) + (t - \sin t) \cdot 0 \right.$$

$$\frac{dy}{dt} = 12 \sin t \quad \left| \quad \frac{dx}{dt} = 10(1 - \cos t) \right.$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{12 \sin t}{10(1 - \cos t)} = \frac{6 \sin t/2}{5(2 \sin^2 t/2)} = \frac{6 \sin \frac{t}{2} \cdot \cos \frac{t}{2}}{5(x) \cdot \sin^2 \frac{t}{2}}$$

$$= \frac{6}{5} \cot \frac{t}{2}$$

MisEx13] $\frac{dy}{dx} = 9$ if $y = \sin^{-1} x + \sin^{-1} \sqrt{1-x^2}$, $0 < x < 1$

put $x = \sin \theta$, $\theta = \sin^{-1} x$

$$y = \sin^{-1}(\sin \theta) + \sin^{-1}(\sqrt{1 - \sin^2 \theta})$$

$$= \theta + \sin^{-1}(\cos \theta)$$

$$= \theta + \sin^{-1}[\sin(\frac{\pi}{2} - \theta)]$$

$$= \theta + \frac{\pi}{2} - \theta$$

$$\frac{dy}{dx} = \frac{2}{1} = 2$$

NOTE: $\sin^{-1} \sqrt{1-x^2} = \cos^{-1} x$

$$\cos^{-1} \sqrt{1-x^2} = \sin^{-1} x$$

$$\therefore y = \sin^{-1} x + \sin^{-1} \sqrt{1-x^2} = \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$$

MisEx14] If $x\sqrt{1+y} + y\sqrt{1+x} = 0$ for $-1 < x < 1$, prove $\frac{dy}{dx} = \frac{-1}{(1+x)^2}$

$$x\sqrt{1+y} + y\sqrt{1+x} = 0$$

$$x\sqrt{1+y} = -y\sqrt{1+x}$$

$$\frac{x}{\sqrt{1+x}} = \frac{-y}{\sqrt{1+y}}$$

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Squaring on both sides,

$$x^2(1+y) = y^2(1+x)$$

$$x^2 + x^2y = y^2 + xy^2$$

$$(x^2 - y^2) + (x^2y - xy^2) = 0$$

$$(x-y)(x+y) + xy(x-y) = 0$$

$$(x-y)(x+y+xy) = 0$$

$$(x-y)(x+y+xy) = 0 \text{ but } x+y \neq 0$$

$$x+y+xy = 0$$

$$y(1+x) = -x$$

$$y = \frac{-x}{1+x}$$

$$\frac{dy}{dx} = \frac{(1+x)(-1) + x}{(1+x)^2} = \frac{-1-x+x}{(1+x)^2} = \frac{-1}{(1+x)^2}$$

Mis Ex 15] If $(x-a)^2 + (y-b)^2 = c^2$, for some $c > 0$. prove
 $\frac{d^2y}{dx^2} \left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{\frac{3}{2}}$ is a constant independent of a & b .

Sol] Gn: $(x-a)^2 + (y-b)^2 = c^2$

$$2(x-a) + 2(y-b)y' = 0$$

$$y' = \frac{-2(x-a)}{2(y-b)} = \frac{-(x-a)}{(y-b)}$$

$$y'' = \frac{(y-b)(-1) + (x-a)y'}{(y-b)^2} = \frac{-(y-b) + (x-a)\left(\frac{-(x-a)}{(y-b)}\right)}{(y-b)^2}$$

$$= \frac{-(y-b)^2 - (x-a)^2}{(y-b)^3} = \frac{-[(y-b)^2 + (x-a)^2]}{(y-b)^3}$$

$$\text{Gn: } \frac{[1 + (y')^2]^{\frac{3}{2}}}{y''} = \frac{\left[1 + \left(\frac{-(x-a)}{(y-b)} \right)^2 \right]^{\frac{3}{2}}}{\frac{-[(y-b)^2 + (x-a)^2]}{(y-b)^3}} = \frac{[(y-b)^2 + (x-a)^2]^{\frac{3}{2}}}{-[(y-b)^2 + (x-a)^2]} = \frac{1}{-1} = -1$$

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$$\begin{aligned}
 &= \left[\frac{(y-b)^2 + (x-a)^2}{(y-b)^2} \right]^{\frac{3}{2}} \cdot \frac{(y-b)^3}{-[(y-b)^2 + (x-a)^2]} \\
 &= \frac{(c^2)^{\frac{3}{2}}}{((y-b)^2)^{\frac{3}{2}}} \cdot \frac{(y-b)^3}{-c^2} = \frac{c^3}{(y-b)^3} \cdot \frac{(y-b)^3}{-c^2} \\
 &= \frac{c^3}{-c^2} = -c \\
 &\therefore \frac{[1+(y')^2]^{\frac{3}{2}}}{y''} \text{ is a constant independent of } a \text{ and } b
 \end{aligned}$$

EXERCISE 5.6

7] $x = \frac{\sin^3 t}{\sqrt{\cos 2t}}$, $y = \frac{\cos^3 t}{\sqrt{\cos 2t}}$. Find $\frac{dy}{dx}$

$$\frac{dx}{dt} = \frac{\sqrt{\cos 2t} \cdot 3\sin^2 t \cdot \cos t + \sin^3 t \cdot \frac{1}{2} \frac{\sin 2t}{\sqrt{\cos 2t}}}{(\sqrt{\cos 2t})^2}$$

$$= \frac{\cos 2t \cdot 3\sin^2 t \cdot \cos t + \sin^3 t \cdot \sin 2t}{(\sqrt{\cos 2t})^3}$$

$$\frac{dy}{dt} = \frac{-\sqrt{\cos 2t} \cdot 3\cos^2 t \cdot \sin t + \cos^3 t \cdot \frac{1}{2} \frac{\sin 2t}{\sqrt{\cos 2t}}}{(\sqrt{\cos 2t})^2}$$

$$= \frac{-\cos 2t \cdot 3\cos^2 t \cdot \sin t + \cos^3 t \cdot \sin 2t}{(\sqrt{\cos 2t})^3}$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{-\cos 2t \cdot 3\cos^2 t \cdot \sin t + \cos^3 t \cdot \sin 2t}{\cos 2t \cdot 3\sin^2 t \cdot \cos t + \sin^3 t \cdot \sin 2t}$$

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$$\frac{dy}{dx} = \frac{-\cos 2t \cdot 3\cos^2 t \cdot \sin t + \cos^3 t \cdot \sin 2t}{(\sqrt{\cos 2t})^3}$$

$$\frac{\cos 2t \cdot 3\sin^2 t \cdot \cos t + \sin^3 t \cdot \sin 2t}{(\sqrt{\cos 2t})^3}$$

$$= \frac{-\cos 2t \cdot 3\cos^2 t \cdot \sin t + \cos^3 t \cdot \sin 2t}{\cos 2t \cdot 3\sin^2 t \cdot \cos t + \sin^3 t \cdot \sin 2t}$$

$$= \frac{-(2\cos^2 t - 1) \cdot 3\cos^2 t \cdot \sin t + \cos^3 t \cdot 2\sin t \cos t}{(1 - 2\sin^2 t) 3\sin^2 t \cdot \cos t + \sin^3 t \cdot 2\sin t \cos t}$$

$$= \frac{-6\cos^4 t \sin t + 3\cos^2 t \sin t + 2\cos^4 t \sin t}{3\sin^2 t \cos t - 6\sin^4 t \cos t + 2\sin^4 t \cos t}$$

$$= \frac{3\cos^2 t \sin t - 4\cos^4 t \sin t}{3\sin^2 t \cos t - 4\sin^4 t \cos t}$$

$$= \frac{\cos t \cdot \sin t (3\cos^2 t - 4\cos^4 t)}{\sin t \cdot \cos t (3\sin^2 t - 4\sin^4 t)}$$

$$= \frac{3\cos^2 t - 4\cos^4 t}{3\sin^2 t - 4\sin^4 t}$$

$$= \frac{-(4\cos^4 t - 3\cos^2 t)}{3\sin^2 t - 4\sin^4 t} = \frac{-\cos 3t}{\sin 3t}$$

$$= -\cot 3t$$

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Continuity:

A function $f(x)$ is said to be continuous if $\lim_{x \rightarrow a} f(x) = f(a)$
 i.e., $LHL = f(a) = RHL$

EXERCISE 5.1

- 17] Find the relation b/w a & b so that the funⁿ f defined by
 $f(x) = \begin{cases} ax+1, & \text{if } x \leq 3 \\ bx+3, & \text{if } x > 3 \end{cases}$ is continuous at $x=3$

Gn: function is continuous at $x=3$

$$\Rightarrow \lim_{x \rightarrow 3^-} f(x) = f(3) = \lim_{x \rightarrow 3^+} f(x)$$

$$\lim_{x \rightarrow 3^-} (ax+1) = \lim_{x \rightarrow 3^+} (bx+3)$$

$$3a+1 = 3b+3$$

$$3a = 3b+2$$

$$a = \frac{3b+2}{3}$$

- 26] $f(x) = \begin{cases} \frac{k \cos x}{\pi - 2x}, & \text{if } x \neq \frac{\pi}{2} \\ 3, & \text{if } x = \frac{\pi}{2} \end{cases}$ is continuous at $x = \frac{\pi}{2}$. $k = ?$

Gn: funⁿ is continuous at $x = \frac{\pi}{2}$

$$\Rightarrow \lim_{x \rightarrow \frac{\pi}{2}} f(x) = f\left(\frac{\pi}{2}\right)$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \left[\frac{k \cos x}{\pi - 2x} \right] = 3$$

$$\lim_{\theta \rightarrow 0} \left[\frac{k \cos\left(\frac{\pi}{2} - \theta\right)}{\theta} \right] = 3$$

let $\pi - 2x = 0$
 substituting x
 $\pi - 0 = 2x$
 $x = \frac{\pi - 0}{2} = \frac{\pi}{2}$
 as $x \rightarrow \frac{\pi}{2}$
 $\pi - 2\left(\frac{\pi}{2}\right) = 0$
 $0 = 0$
 $\theta \rightarrow 0$

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$$\lim_{\theta \rightarrow 0} \left(\frac{k \sin \frac{\theta}{2}}{2 \cdot \frac{\theta}{2}} \right) = 3$$

$$\frac{k}{2} \cdot 1 = 3$$

$$\therefore k = 6$$

27] $f(x) = \begin{cases} kx^2, & \text{if } x \leq 2 \\ 3, & \text{if } x > 2 \end{cases}$ is continuous at $x = 2$, $k = ?$

Given function is continuous at $x = 2$

$$\Rightarrow \lim_{x \rightarrow 2^-} f(x) = f(2) = \lim_{x \rightarrow 2^+} f(x)$$

$$\lim_{x \rightarrow 2^-} (kx^2) = \lim_{x \rightarrow 2^+} (3)$$

$$4k = 3$$

$$k = \frac{3}{4}$$

28] $f(x) = \begin{cases} kx+1, & \text{if } x \leq \pi \\ \cos x, & \text{if } x > \pi \end{cases}$ is continuous at $x = \pi$, $k = ?$

Given $f(x)$ is continuous at $x = \pi$

$$\Rightarrow \lim_{x \rightarrow \pi^-} f(x) = f(\pi) = \lim_{x \rightarrow \pi^+} f(x)$$

$$\lim_{x \rightarrow \pi^-} (kx+1) = \lim_{x \rightarrow \pi^+} (\cos x)$$

$$\pi k + 1 = +\cos \pi = -1$$

$$k = \frac{-2}{\pi}$$

** 29] $f(x) = \begin{cases} kx+1, & \text{if } x \leq 5 \\ 3x-5, & \text{if } x > 5 \end{cases}$ is continuous at $x = 5$, $k = ?$

$$\Rightarrow \lim_{x \rightarrow 5^-} f(x) = f(5) = \lim_{x \rightarrow 5^+} f(x)$$

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$$\lim_{x \rightarrow 5^+} (kx+1) = \lim_{x \rightarrow 5^+} (3x-5)$$

$$5k+1 = 15-5$$

$$5k = 9$$

$$k = \frac{9}{5}$$

- 30] Find a, b such that the funⁿ defined by $f(x) = \begin{cases} 5, & \text{if } x \leq 2 \\ ax+b, & \text{if } 2 < x < 10 \\ 21, & \text{if } x \geq 10 \end{cases}$ is a continuous funⁿ.

On funⁿ is continuous

at $x=2$

$$\lim_{x \rightarrow 2^-} f(x) = f(2) = \lim_{x \rightarrow 2^+} f(x)$$

$$\lim_{x \rightarrow 10^-} f(x) = f(10) = \lim_{x \rightarrow 10^+} f(x)$$

$$5 = \lim_{x \rightarrow 2^+} (ax+b)$$

$$\lim_{x \rightarrow 10^-} (ax+b) = 21$$

$$10a+b = 21 \rightarrow (2)$$

$$5 = 2a+b \rightarrow (1)$$

$$(2) - (1) \Rightarrow$$

$$10a+b = 21$$

$$- 2a+b = 5$$

$$8a = 16$$

$$\boxed{a=2}$$

$$(1) \Rightarrow 4+b = 5$$

$$\boxed{b=1}$$

- 18] For what value of λ is the funⁿ defined by $f(x) = \begin{cases} \lambda(x^2-2x) & \text{if } x \leq 0 \\ 4x+1 & \text{if } x > 0 \end{cases}$ is continuous at $x=0$. What about continuity at $x=1$.

On funⁿ is continuous at $x=0$

$$\lim_{x \rightarrow 0^-} f(x) = f(0) = \lim_{x \rightarrow 0^+} f(x)$$

$$\lim_{x \rightarrow 0^-} [\lambda(x^2-2x)] = \lim_{x \rightarrow 0^+} (4x+1)$$

$$\lambda(0) = 1 \Rightarrow \lambda \text{ vanishes for any value of } \lambda$$

$\therefore$ For no value of λ $f(x)$ is continuous at $x=0$.

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$$\text{At } x=1: \lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} (4x+1) = 5$$

at $x=1$, $f(x)$ is a constant at any value of x it is continuous.

- 1] Prove that the funⁿ $f(x) = 5x-3$ is continuous i) at $x=0$
 ii) at $x=-3$
 and iii) at $x=5$

i) at $x=0$,

$$f(0) = 5(0) - 3 = -3$$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} (5x-3) = 5(0) - 3 = -3$$

$$\therefore \lim_{x \rightarrow 0} f(x) = f(0) \Rightarrow \text{fun}^n \text{ is cont}^n \text{ at } x=0$$

ii) at $x=-3$

$$f(-3) = 5(-3) - 3 = -18$$

$$\lim_{x \rightarrow -3} f(x) = \lim_{x \rightarrow -3} (5x-3) = 5(-3) - 3 = -18$$

$$\therefore \lim_{x \rightarrow -3} f(x) = f(-3) \Rightarrow \text{fun}^n \text{ is cont}^n \text{ at } x=-3$$

iii) at $x=5$

$$f(5) = 5(5) - 3 = 22$$

$$\lim_{x \rightarrow 5} f(x) = \lim_{x \rightarrow 5} (5x-3) = 5(5) - 3 = 22$$

$$\therefore \lim_{x \rightarrow 5} f(x) = f(5) \Rightarrow \text{fun}^n \text{ is cont}^n \text{ at } x=5$$

- 2] Examine the continuity of the funⁿ $f(x) = 2x^2-1$ at $x=3$.

at $x=3$; ~~$f(3)$~~ $\lim_{x \rightarrow 3} f(x) = f(3)$

Consider, $x \rightarrow 3$

$$\lim_{x \rightarrow 3} (2x^2-1) = 2(3)^2-1$$

$$2(3^2)-1 = 2(3)^2-1$$

$$17 = 17$$

$$\therefore \text{fun}^n \text{ is cont}^n \text{ at } x=3$$

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5] Is the funⁿ f defined by $f(x) = \begin{cases} x, & \text{if } x \leq 1 \\ 5, & \text{if } x > 1 \end{cases}$ continuous at $x=0$? at $x=1$? at $x=2$?

At $x=1$, $\lim_{x \rightarrow 1^-} f(x) = f(1) = \lim_{x \rightarrow 1^+} f(x)$

$$\lim_{x \rightarrow 1^-} (x) = 1$$

$$\text{but } 1 \neq 5$$

$\therefore$ funⁿ is not contⁿ at $x=1$ and hence it is contⁿ at all other points.

$$\text{At } x=0, \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} (x) = 0$$

$\Rightarrow f(x)$ is contⁿ at $x=0$

$$\text{At } x=2, \lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} (5) = 5$$

$\Rightarrow f(x)$ is contⁿ at $x=2$

Find all points of discontinuity of f , where f is defined by

$$6] f(x) = \begin{cases} 2x+3, & \text{if } x \leq 2 \\ 2x-3, & \text{if } x > 2 \end{cases}$$

$$\text{At } x=2, \lim_{x \rightarrow 2^-} f(x) = f(2) = \lim_{x \rightarrow 2^+} f(x)$$

$$\lim_{x \rightarrow 2^-} (2x+3) = \lim_{x \rightarrow 2^+} (2x-3)$$

$$\text{but } 4+3 \neq 4-3$$

$$7 \neq 1$$

LHL is not equal to RHL

$\therefore$ The given funⁿ is not contⁿ at $x=2$

Point of discontinuity is $x=2$

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$$7) f(x) = \begin{cases} (x)+3, & \text{if } x \leq -3 \\ -2x, & \text{if } -3 < x < 3 \\ 6x+2, & \text{if } x \geq 3 \end{cases}$$

(a) At $x = -3$: $\lim_{x \rightarrow -3^-} f(x) = f(-3) = \lim_{x \rightarrow -3^+} f(x)$

$$\lim_{x \rightarrow -3^-} f(x) = f(-3) = \lim_{x \rightarrow -3^+} f(x)$$

$$\lim_{x \rightarrow -3^-} (x+3) = \lim_{x \rightarrow -3^+} (-2x)$$

$$-3+3 = -2(-3)$$

$$0 = 6$$

 $\therefore$ The given funⁿ is contⁿ at $x = -3$ (b) At $x = 3$:

$$\lim_{x \rightarrow 3^-} f(x) = f(3) = \lim_{x \rightarrow 3^+} f(x)$$

$$\lim_{x \rightarrow 3^-} (-2x) = \lim_{x \rightarrow 3^+} (6x+2)$$

$$-6 = 20$$

 $\therefore$ The given funⁿ is not continuous at $x = 3$ $\therefore$ The point is discontinuous at $x = 3$

$$8) f(x) = \begin{cases} \frac{|x|}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

$$\text{LHL} = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{|x|}{x} = \lim_{x \rightarrow 0^-} \frac{-x}{x} = \lim_{x \rightarrow 0^-} (-1) = -1$$

$$f(0) = 0$$

$$\text{RHL} = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{|x|}{x} = \lim_{x \rightarrow 0^+} \frac{x}{x} = \lim_{x \rightarrow 0^+} (1) = 1$$

LHL $\neq$ RHL, $f(x)$ is not contⁿ $\therefore$ pt. of discontinuity is $x = 0$

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9) $f(x) = \begin{cases} \frac{x}{|x|}, & x < 0 \\ -1, & x \geq 0 \end{cases}$

LHL: $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{x}{|x|} = \lim_{x \rightarrow 0^-} \frac{x}{-x} = \lim_{x \rightarrow 0^-} -1 = -1$

$f(0) = -1$

RHL: $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (-1) = -1$

$\therefore$ LHL = RHL, $f(x)$ is continuous.

$\therefore$ no point of discontinuity.

10) $f(x) = \begin{cases} x+1, & x \geq 1 \\ x^2+1, & x < 1 \end{cases}$

LHL: $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x^2+1) = (1^2+1) = 2$

$f(1) = 1+1 = 2$

RHL: $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (x+1) = 1+1 = 2$

$\therefore$ LHL = RHL, $f(x)$ is continuous.

$\therefore$ no point of discontinuity.

11) $f(x) = \begin{cases} x^3-3, & x \leq 2 \\ x^2+1, & x > 2 \end{cases}$

$\lim_{x \rightarrow 2^-} f(x) = f(2) = \lim_{x \rightarrow 2^-} (x^3-3)$

$\lim_{x \rightarrow 2^-} (x^3-3) = \lim_{x \rightarrow 2^-} (x^2+1) = 4+1 = 5$

$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (x^2+1) = 4+1 = 5$

$\therefore f(x)$ is continuous at $x = 2$

There is no pt. of discontinuity.

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$$12] f(x) = \begin{cases} x^{10}-1, & x \leq 1 \\ x^2, & x > 1 \end{cases}$$

$$LHL = \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x^{10}-1) = (1^{10}-1) = 0$$

$$f(1) = 1^{10}-1 = 0$$

$$RHL = \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (x^2) = 1^2 = 1$$

$LHL \neq RHL \therefore f(x)$ is not contⁿ at $x=1$.
 $\therefore$ pt. of discontinuity is $x=1$.

$$13] \text{ If the funⁿ defined by } f(x) = \begin{cases} x+5, & x \leq 1 \\ x-5, & x > 1 \end{cases} \text{ is a contⁿ fun}$$

$$\lim_{x \rightarrow 1^-} f(x) = f(1) = \lim_{x \rightarrow 1^+} f(x)$$

$$\lim_{x \rightarrow 1^-} (x+5) \neq \lim_{x \rightarrow 1^+} (x-5)$$

$$1+5 \neq 1-5$$

$\therefore$ It is not continuous.

$$14] \text{ Discuss the continuity of funⁿ } f \text{ where } f \text{ is defined by}$$

$$f(x) = \begin{cases} 3, & 0 \leq x \leq 1 \\ 4, & 1 < x < 3 \\ 5, & 3 \leq x \leq 10 \end{cases}$$

$$\text{At } x=1: \lim_{x \rightarrow 1^-} f(x) = f(1) = \lim_{x \rightarrow 1^+} f(x)$$

$$\lim_{x \rightarrow 1^-} (3) = (3) = \lim_{x \rightarrow 1^+} (4)$$

$$3 \neq 4$$

$\therefore f(x)$ is discontinuous at $x=1$.

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At $x=3$: $\lim_{x \rightarrow 3^-} f(x) = f(3) = \lim_{x \rightarrow 3^+} f(x)$

$$0 = (1 - \lim_{x \rightarrow 3^-} (4)) = \lim_{x \rightarrow 3^+} (5) = 1$$

$$4 \neq 5$$

$\therefore f(x)$ is discontinuous at $x=3$

15) $f(x) = \begin{cases} 2x, & x < 0 \\ 0, & 0 \leq x \leq 1 \\ 4x, & x > 1 \end{cases}$

At $x=0$: $\lim_{x \rightarrow 0^-} f(x) = f(0) = \lim_{x \rightarrow 0^+} f(x)$

$$\lim_{x \rightarrow 0^-} (2x) = \lim_{x \rightarrow 0^+} (0)$$

$$2(0) = 0 \quad (2+x) \lim_{x \rightarrow 0^-} = 0$$

$$0 = 0$$

$\therefore f(x)$ is contⁿ at $x=0$

At $x=1$: $\lim_{x \rightarrow 1^-} f(x) = f(1) = \lim_{x \rightarrow 1^+} f(x)$

$$\lim_{x \rightarrow 1^-} (0) = \lim_{x \rightarrow 1^+} (4x)$$

$$0 \neq 4(1) = 4$$

$$0 \neq 4x > 1$$

$\therefore f(x)$ is not contⁿ at $x=1$

25) Examine the continuity of f where f is defined by

$$f(x) = \begin{cases} \sin x - \cos x, & x \neq 0 \\ -1, & x = 0 \end{cases}$$

$\lim_{x \rightarrow 0^-} f(x) = f(0) = \lim_{x \rightarrow 0^+} f(x)$

$$\lim_{x \rightarrow 0^-} (\sin x - \cos x) = -1 = \lim_{x \rightarrow 0^+} (\sin x - \cos x)$$

$$(0 - 1) = -1 = (0 - 1)$$

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$$f(-1) = -1$$

$\therefore \text{LHL} = \text{RHL}, f(x)$ is continuous.

$$16) f(x) = \begin{cases} -2, & x \leq -1 \\ 2x, & -1 \leq x \leq 1 \\ 2, & x > 1 \end{cases}$$

$$\text{At } x = -1: \lim_{x \rightarrow -1^-} f(x) = f(-1) = \lim_{x \rightarrow -1^+} f(x)$$

$$\lim_{x \rightarrow -1^-} (-2) = \lim_{x \rightarrow -1^+} (2x)$$

$$-2 = -2$$

$\therefore f(x)$ is contⁿ at $x = -1$.

$$\text{At } x = 1: \lim_{x \rightarrow 1^-} f(x) = f(1) = \lim_{x \rightarrow 1^+} f(x)$$

$$\lim_{x \rightarrow 1^-} (2x) = \lim_{x \rightarrow 1^+} (2)$$

$$2(1) = 2$$

$$2 = 2$$

$\therefore f(x)$ is contⁿ at $x = 1$.

Algebra of Continuous functions:

Theorem 1: Suppose f and g are two real funⁿs continuous at a real no. c , then

(a) $f + g$ is contⁿ at $x = c$

(b) $f - g$ is contⁿ at $x = c$

(c) $f \cdot g$ is contⁿ at $x = c$

(d) $\frac{f}{g}$ is contⁿ at $x = c$; $g(x) \neq 0$

Theorem 2: Suppose f and g are real valued funⁿs such that $(f \circ g)$ is defined at c . If g and f are both contⁿ at c , then their composition $(f \circ g)$ is also contⁿ at c .

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Eg 19] Show that the funⁿ $f(x) = \sin(x^2)$ is a contⁿ funⁿ.
 Let $f(x) = \sin(x^2)$,
 $g(x) = \sin x$, $h(x) = x^2$
 $g(x)$ being trigonometric is contⁿ funⁿ
 $h(x)$ being polynomial is a contⁿ funⁿ
 Their composition: $g \circ h(x) = g[h(x)] = g(x^2) = \sin(x^2) = f(x)$
 $\therefore g$ and h are contⁿ funⁿs, their composition $g \circ h(x) = f(x)$ is also contⁿ.

31] Show that funⁿ $f(x) = \cos(x^2)$ is a contⁿ funⁿ.
 Let $f(x) = \cos(x^2)$,
 $g(x) = \cos x$, $h(x) = x^2$
 $g(x)$ being trigonometric is contⁿ funⁿ
 $h(x)$ being polynomial is a contⁿ funⁿ
 Their composition: $g \circ h(x) = g[h(x)] = g(x^2) = \cos(x^2) = f(x)$
 $\therefore g$ and h are contⁿ funⁿs, their composition $g \circ h(x) = f(x)$ is also contⁿ.

32] Show that $f(x) = |\cos x|$ is a contⁿ funⁿ.
 Gn: $f(x) = |\cos x|$
 let $g(x) = |x|$, $h(x) = \cos x$
 their composition $g \circ h(x) = g[h(x)] = g(\cos x) = |\cos x| = f(x)$
 $g(x)$ being modulus funⁿ is contⁿ funⁿ.
 $h(x)$ being trigonometric funⁿ is contⁿ funⁿ.
 $\therefore g$ and h are contⁿ funⁿs, their composition $g \circ h(x) = f(x)$ is also contⁿ.

33] Examine that $\sin|x|$ is a contⁿ funⁿ.

let $f(x) = \sin|x|$
 let $g(x) = \sin x$, $h(x) = |x|$

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Their composition: $g \circ h(x) = g(h(x)) = g(\sin x) = \sin |x| = f(x)$
 $g(x)$ being trigonometric funⁿ is contⁿ
 $h(x)$ being modulus funⁿ is contⁿ
 $\therefore g$ and h are contⁿ funⁿ, their composition $g \circ h(x) = f(x)$ is also contⁿ.

34] Find all pts of discontinuity of f , $f(x) = |x| - |x+1|$:

Sol] Let $g(x) = |x|$
 $h(x) = x+1$

$$g \circ h(x) = g(h(x)) = g(x+1) = |x+1|$$

$|x|$ and $x+1$ both being continuous functions on $\mathbb{R}$,
 $g \circ h(x)$ is also continuous function on $\mathbb{R}$.

Also $|x|$ is continuous function.

$\therefore$ Their difference, $|x| - |x+1|$ is also continuous on $\mathbb{R}$.

$\Rightarrow f(x)$ is continuous on $\mathbb{R}$.

$\Rightarrow f$ has no point of discontinuity.

Eg 20] Show that $f(x) = |1-x+|x||$ where x is any real no. is a contⁿ funⁿ.

$$\text{Ans: } f(x) = |1-x+|x||$$

$$\text{let, } g(x) = |x|, h(x) = 1-x+|x|$$

$$\text{Their composition: } g \circ h(x) = g(h(x)) = g(1-x+|x|) = |1-x+|x|| = f(x)$$

$g(x)$ being modulus funⁿ is contⁿ

$h(x)$ being sum of a polynomial and a modulus funⁿ is contⁿ.

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∴ Their composition go $h(x) = f(g(x))$ is also contⁿ.

Mean Value Theorems:

Theorem 6: ROLLE'S Theorem, (RT)

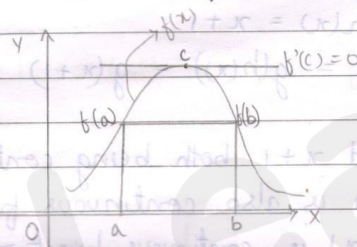
Let f be a funⁿ $f: [a, b] \rightarrow \mathbb{R}$ be contⁿ on $[a, b]$

(ii) differentiable on (a, b)

(iii) $f(a) = f(b)$ where a and b are some real no.s,

then there exists atleast one pt. c such that

$$f'(c) = 0$$



(LMVT)

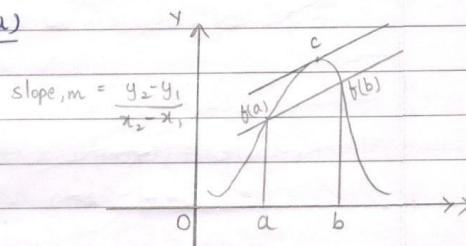
Theorem 7: Let a funⁿ $f: [a, b] \rightarrow \mathbb{R}$ be

i) be contⁿ $[a, b]$

ii) differentiable in (a, b) , then

there exist atleast one pt. c such that,

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$



EXERCISE 5.8

1) Verify Rolle's theorem for the funⁿ $f(x) = x^2 + 2x - 8$; $x \in [-4, 2]$

Given: $f(x) = x^2 + 2x - 8$

(1) $f(x)$ being polynomial is contⁿ in $[-4, 2]$

(2) $f'(x) = 2x + 2 \Rightarrow f(x)$ is differentiable in $(-4, 2)$

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$$(3) a = -4, f(a) = f(-4) = 16 - 8 - 8 = 0$$

$$b = 2, f(b) = f(2) = 4 + 4 - 8 = 0$$

$$f(a) = f(b)$$

$\therefore$ all 3 conditions of RT are satisfied, there exists atleast one pt. c such that

$$f'(c) = 0$$

$$2c + 2 = 0$$

$$2c = -2$$

$$c = -1 \in (-4, 2)$$

Hence, RT verified.

2) Examine if Rolle's theorem is applicable, if so verify

i) $f(x) = x^2 - 1$ for $x \in [1, 2]$

Gn: $f(x) = x^2 - 1$

(1) $f(x)$ being polynomial is contⁿ in $[1, 2]$

(2) $f'(x) = 2x \Rightarrow f(x)$ is differentiable in $(1, 2)$

(3) $a = 1, f(a) = f(1) = 1^2 - 1 = 0$

$b = 2, f(b) = f(2) = 4 - 1 = 3$

$f(a) \neq f(b)$

$\therefore$ RT is not applicable.

ii) $f(x) = [x], x \in [5, 9]$

Gn: $f(x) = [x]$ is not a contⁿ funⁿ, here it is discontinuous at the integral pt's 5, 6, 7, 8, 9

Hence, RT is not applicable.

*** iii) $f(x) = [x], x \in [-2, 2]$

Gn: $f(x) = [x]$ is not a contⁿ funⁿ, here it is discontinuous at the integral pt's -2, -1, 0, 1, 2

Hence, RT is not applicable.

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Eg 42] Verify RT for $y = x^2 + 2$, $a = -2$, $b = 2$ Gn: $f(x) = x^2 + 2$, $x \in [-2, 2]$ (1) $f(x)$ being polynomial is contn in $[-2, 2]$ (2) $f'(x) = 2x \Rightarrow f(x)$ is differentiable in $(-2, 2)$ (3) $a = -2$, $f(a) = f(-2) = (-2)^2 + 2 = 4 + 2 = 6$ $b = 2$, $f(b) = f(2) = 2^2 + 2 = 4 + 2 = 6$ $\therefore f(a) = f(b)$ $\therefore$ all 3 conditions of RT are satisfied, there exists atleast one pt. c such that

$$f'(c) = 0$$

$$2c = 0$$

$$c = 0, c \in (-2, 2)$$

Hence, RT verified.

Eg 43] Verify MVT for $f(x) = x^2$, $[2, 4]$ Gn: $f(x) = x^2$ (1) $f(x)$ being polynomial is contn in $[2, 4]$ (2) $f'(x) = 2x \Rightarrow f(x)$ is differentiable in $(2, 4)$ $\therefore$ 2 conditions of MVT are satisfied, there exists atleast one pt. c such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$f(a) = f(2) = 4$$

$$f(b) = f(4) = 16$$

$$b - a = 4 - 2 = 2$$

$$2c = \frac{16 - 4}{2}$$

$$2c = \frac{12}{2} = 6$$

$$c = 3 \in (2, 4)$$

Hence, MVT verified.

4] Verify MVT if $f(x) = x^2 - 4x + 3$, $x \in [a, b]$ where $a = 1$, $b = 4$.

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$$\text{Gn: } f(x) = x^2 - 4x - 3 \quad x \in [1, 4]$$

(1) $f(x)$ being polynomial is contⁿ in $[1, 4]$

(2) $f'(x) = 2x - 4 \Rightarrow f(x)$ is differentiable in $(1, 4)$

$\therefore$ 2 conditions of MVT are satisfied, there exists atleast one pt. c such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$f(a) = f(1) = 1 - 4 - 3 = -6$$

$$f(b) = f(4) = 16 - 16 - 3 = -3$$

$$2c - 4 = \frac{-3 - (-6)}{4 - 1} = \frac{-3 + 6}{3} = \frac{3}{3} = 1$$

$$2c - 4 = 1$$

$$2c = 5$$

$$\frac{5}{2} = c = 2.5 \in (1, 4)$$

$\therefore$ MVT verified.

*5] Verify MVT if $f(x) = x^3 - 5x^2 - 3x$, $x \in [a, b]$ where $a=1$, $b=3$. Find all $c \in (1, 3)$ for all $f'(c) = 0$.

$$\text{Gn: } f(x) = x^3 - 5x^2 - 3x$$

(a) (1) $f(x)$ being polynomial is contⁿ in $[1, 3]$.

(2) $f'(x) = 3x^2 - 10x - 3 \Rightarrow f(x)$ is differentiable in $(1, 3)$

$\therefore$ 2 conditions of MVT are satisfied, there exists atleast one pt. c such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$f(a) = f(1) = 1 - 5 - 3 = -7$$

$$f(b) = f(3) = 27 - 45 - 9 = -27$$

$$b - a = 3 - 1 = 2$$

$$3c^2 - 10c - 3 = \frac{-27 - (-7)}{2} = \frac{-27 + 7}{2} = \frac{-20}{2} = -10$$

$$3c^2 - 10c - 3 = -10$$

$$3c^2 - 10c + 7 = 0$$

$$3c^2 - 10c - 3 = -10$$

$$3c^2 - 10c + 7 = 0$$

$$3c^2 - 3c - 7c + 7 = 0$$

$$3c(c-1) - 7(c-1) = 0$$

$$(3c-7), (c-1)$$

$$c = \frac{7}{3} = 2.33 \in (1, 3)$$

$$c = 1 \notin (1, 3)$$

$$\therefore c = 2.33 \in (1, 3)$$

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$$(b) f'(c) = 0 \Rightarrow [1, 3] \rightarrow x \in \mathbb{R} \Rightarrow x^2 - 10x + 3 = 0 \Rightarrow (x-1)(x-3) = 0$$

$$3c^2 - 10c - 3 = 0$$

$$c = \frac{10 \pm \sqrt{100 + 36}}{6}$$

$$c = 3.6, -0.27$$

$$c_1 = 3.6 \notin (1, 3)$$

$$c_2 = -0.27 \notin (1, 3)$$

$\therefore$ There exists no value of $c \in (1, 3)$ when $f'(c) = 0$

(b) Find K if $f(x) = \begin{cases} \frac{1 - \cos 2x}{1 + \cos x}, & x \neq 0 \\ K, & x = 0 \end{cases}$ is continuous @ $x = 0$

Given: $f(x)$ is continuous at $x = 0$
 $\Rightarrow \lim_{x \rightarrow 0} f(x) = f(0)$
 $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{1 + \cos x} = K$
 $\lim_{x \rightarrow 0} \frac{2 \sin^2 x}{1 + \cos x} = K$
 $\lim_{x \rightarrow 0} \left(\frac{\sin^2 x}{x^2} \cdot \frac{x^2}{1 + \cos x} \right) = K$
 $\lim_{x \rightarrow 0} \left(\frac{\sin^2(\frac{\pi}{2})}{\frac{\pi^2}{4}} \cdot \frac{\frac{\pi^2}{4}}{1 + \cos(\frac{\pi}{2})} \right) = K$
 $\frac{1}{1} \cdot 4 = K$
 $\boxed{K = 4}$

(a) PT $\int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_a^{2a} f(x) dx$; if $f(2a-x) = f(x)$
 0; if $f(2a-x) = -f(x)$

and hence evaluate $\int_0^{2\pi} \cos^5 x dx$

In LHS: In 2nd integral: $x = 2a - t$ | $x = a, t = a$
 $dx = -dt$ | $x = 2a, t = 0$

$I = \int_0^a f(x) dx + \int_a^{2a} f(x) dx$
 $= \int_0^a f(x) dx - \int_a^0 f(2a-t) dt = \int_0^a f(x) dx + \int_0^a f(2a-t) dt$
 $\therefore \int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a-x) dx$

case (i): $f(2a-x) = f(x)$
 $\int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(x) dx = 2 \int_0^a f(x) dx$

case (ii): $f(2a-x) = -f(x)$
 $\int_0^{2a} f(x) dx = \int_0^a f(x) dx - \int_0^a f(x) dx = 0$

(b) PT $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a^2 + b^2 + ca)$

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RHS = $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix}$ $C_2 \rightarrow C_2 - C_1; C_3 \rightarrow C_3 - C_1$
 $= \begin{vmatrix} 1 & 0 & 0 \\ a & b-a & c-a \\ a^3 & b^3-a^3 & c^3-a^3 \end{vmatrix}$